

Saddle-Field Policy Optimization for Zero-Sum Markov Games: Adaptive Lyapunov-Drift Control with Finite-Time Stationarity Guarantees

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Abstract—Two-player zero-sum Markov games model robust reinforcement learning and competitive policy optimization. Their stochastic saddle fields can rotate near stationary points. We propose an online controller that assigns separate step sizes for field and curvature directions. A box-constrained quadratic program (QP) minimizes a model of the one-step change in a composite Lyapunov function $\mathcal{V}$, which combines weighted field energy and a regularized policy-space gap. A symmetric–antisymmetric Jacobian decomposition identifies when the curvature direction decreases field energy. For biased stochastic oracles, we prove an $O(1/K)$ bound on the time-average expected stationarity residual, where K is the number of joint updates, plus a floor caused by oracle and local-model errors. A curvature-improvement condition lower-bounds the extra predicted decrease from the curvature step. If the field direction decreases $\mathcal{V}$ proportionally to its current value, $\mathbb{E}[\mathcal{V}]$ decreases geometrically to an error floor. A policy-space bound converts this result into bounds on the unregularized Nash gap and the maximizing player’s worst-case-return loss. Experiments on normal-form, tabular, neural-policy, and trajectory-sampled games confirm the predicted geometry. Across these tasks, the controller attains consistently higher mean worst-case return and lower mean exploitability than the matched field-only ablation. On neural CyclicControl, it reduces final exploitability by 79.1%.

Index Terms—Zero-sum Markov games, robust reinforcement learning, stochastic saddle fields, Lyapunov drift, adaptive step size.

I. INTRODUCTION

TWO-PLAYER zero-sum Markov games provide a standard framework for competitive policy optimization [1]–[3]. Robust reinforcement learning (RL) applies this framework to policies that must retain performance under model mismatch, disturbances, and adversarial perturbations [4]–[7]. Robust adversarial reinforcement learning (RARL) represents the task and disturbance policies as the maximizing and minimizing players [8], [9].

Policy parameterization couples the two update blocks. Incompatible update scales may induce oscillatory or divergent recursions. Linear-quadratic analysis shows that alternating RARL updates without a stabilizing initialization can destabilize the closed-loop system, whereas a robustly stabilizing initialization and a double-loop update restore convergence guarantees [10].

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Saddle-field viewpoint. For differentiable maximizing- and minimizing-player policy parameters θ and ψ , let $J(\theta, \psi)$ denote the maximizing player’s return and write $z \triangleq (\theta, \psi)$. We express policy ascent and descent as the saddle field

$$F(z) \triangleq \begin{bmatrix} -\nabla_{\theta} J(\theta, \psi) \\ \nabla_{\psi} J(\theta, \psi) \end{bmatrix}. \quad (1)$$

Here $\mathcal{Z}^* \triangleq \{z : F(z) = 0\}$ is the first-order stationary set, and the Jacobian $\nabla F(z)$ is the derivative of F with respect to z . Stochastic policy-gradient estimators provide noisy observations of this field and its Jacobian actions.

Gradient descent–ascent (GDA) can rotate or diverge even on bilinear games. The proximal point method (PPM) and extragradient-type methods, including the classical extragradient method (EGM), Mirror-Prox, and policy extragradient, introduce stabilizing corrections [11]–[15]. For a scalar step size $s > 0$, their representative updates are

$$\begin{aligned} \text{GDA: } z_+^{\text{GDA}} &= z - sF(z), \\ \text{PPM: } z_+^{\text{PPM}} &= z - sF(z_+^{\text{PPM}}), \\ \text{EGM: } \tilde{z} &= z - sF(z), \quad z_+^{\text{EGM}} = z - sF(\tilde{z}), \end{aligned} \quad (2)$$

where $\tilde{z}$ is the EGM extrapolation point. GDA is explicit, PPM is implicit, and EGM uses an extrapolated query to approximate the PPM correction.

The local mechanism follows from the field energy and the curvature-induced direction

$$\phi(z) \triangleq \frac{1}{2} \|F(z)\|^2, \quad G(z) \triangleq \nabla F(z)F(z), \quad (3)$$

where $\|\cdot\|$ is the Euclidean norm. Because $\phi(z) \geq 0$ and $\phi(z) = 0$ exactly when $z \in \mathcal{Z}^*$, field energy measures first-order stationarity. Proposition 3 shows that PPM and EGM locally add $+s^2 G$ to the GDA direction $-sF$. High-resolution ordinary differential equation (ODE) analysis also exposes curvature corrections that distinguish stable minimax updates from the basic field update [16]. Game decompositions separate ∇F into symmetric potential-like and antisymmetric Hamiltonian components [17], [18]. Hamiltonian-gradient methods use $\nabla F^\top F$, whereas symplectic adjustments isolate antisymmetric coupling [19], [20].

Two design questions. Proposition 3 fixes the step sizes of $-F$ and $+G$ to (s, s^2) . A double-step stochastic extragradient method uses an extrapolation step size $\gamma_t > 0$ and a final step size $\eta_t > 0$, giving the local pair $(\eta_t, \eta_t \gamma_t)$ [21]. Adaptive Mirror-Prox selects a common geometry from past gradients [22]; existing Lyapunov rules tune one step size [23], [24].

These methods motivate two questions: When does $+G$ reduce field energy, and how should its step size be balanced with that of $-F$?

We answer both questions with an online controller. It embeds ϕ and a regularized policy-space gap in a composite Lyapunov function $\mathcal{V}$. An exact symmetric–antisymmetric decomposition identifies when $+G$ decreases ϕ . A two-dimensional box-constrained quadratic program (QP) selects both step sizes by minimizing a local model of the one-step change in $\mathcal{V}$.

Contributions. The main contributions are as follows.

- **Curvature geometry and adaptive step sizes.** An exact symmetric–antisymmetric condition characterizes curvature descent. A closed-form box QP selects both step sizes and gives no worse predicted change in $\mathcal{V}$ than feasible one-direction and fixed-coupling rules.
- **Finite-time stochastic guarantees.** For biased stochastic oracles, the time-average expected stationarity residual has an $O(1/K)$ transient, where K is the number of joint updates, plus an explicit error floor. Under local field–Lyapunov alignment, $\mathbb{E}[\mathcal{V}]$ contracts geometrically, and Proposition 1 yields Nash-gap and worst-case-return bounds.
- **Mechanism and policy-game evidence.** Normal-form experiments verify the curvature geometry; population and trajectory-sampled Markov games show higher worst-case return and lower exploitability than the matched field-only ablation.

The rest of the paper is organized as follows. Section II develops the saddle-field geometry. Sections III and IV present the adaptive step-size rule and its stochastic analysis. Sections V and VI give the implementation and experiments.

II. SADDLE-FIELD FORMULATION AND THE GEOMETRY OF THE CURVATURE DIRECTION

A. Parameterized zero-sum Markov games as saddle fields

We consider an infinite-horizon discounted two-player zero-sum Markov game with state space $\mathcal{S}$, maximizing-player action space $\mathcal{A}$, minimizing-player action space $\mathcal{B}$, transition kernel P , bounded reward r to the maximizing player, discount factor $\alpha \in (0, 1)$, and initial-state distribution ρ . Unless stated otherwise, all policy-space suprema and infima below are over stationary randomized Markov policies; for finite discounted zero-sum games this class contains optimal policies. For a maximizing-player policy π and a minimizing-player policy ν , the maximizing player’s discounted return is

$$J(\pi, \nu) \triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t=0}^{\infty} \alpha^t r(s_t, a_t, b_t) \right].$$

Here $\mathbb{E}_{\rho, \pi, \nu}$ denotes expectation under the initial-state distribution and the two policies. The policy-space saddle objective is $\sup_{\pi} \inf_{\nu} J(\pi, \nu)$. Define the maximizing player’s worst-case return, game value, and unregularized policy-space Nash gap by

$$\begin{aligned} R_{\text{BR}}(\pi) &\triangleq \inf_{\bar{\nu}} J(\pi, \bar{\nu}), & v^* &\triangleq \sup_{\bar{\pi}} R_{\text{BR}}(\bar{\pi}), \\ \text{Gap}_0(\pi, \nu) &\triangleq \sup_{\bar{\pi}} J(\bar{\pi}, \nu) - \inf_{\bar{\nu}} J(\pi, \bar{\nu}). \end{aligned} \quad (4)$$

For finite action sets, define the discounted causal entropies

$$\begin{aligned} H_{\alpha}^{\pi}(\pi, \nu) &\triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t \geq 0} \alpha^t (-\log \pi(a_t | s_t)) \right], \\ H_{\alpha}^{\nu}(\pi, \nu) &\triangleq \mathbb{E}_{\rho, \pi, \nu} \left[\sum_{t \geq 0} \alpha^t (-\log \nu(b_t | s_t)) \right], \end{aligned}$$

with the convention $0 \log 0 = 0$. For a regularization strength $\varepsilon \geq 0$, define

$$J_{\varepsilon}(\pi, \nu) \triangleq J(\pi, \nu) + \varepsilon H_{\alpha}^{\pi}(\pi, \nu) - \varepsilon H_{\alpha}^{\nu}(\pi, \nu), \quad (5)$$

with Gap_{ε} defined from J_{ε} as in (4).

Proposition 1 (Policy-space performance bridge). *Suppose $\mathcal{A}$ and $\mathcal{B}$ are finite. For every policy pair (π, ν) and $\varepsilon \geq 0$,*

$$\begin{aligned} 0 &\leq v^* - R_{\text{BR}}(\pi) \leq \text{Gap}_0(\pi, \nu) \\ &\leq \text{Gap}_{\varepsilon}(\pi, \nu) + \frac{\varepsilon (\log |\mathcal{A}| + \log |\mathcal{B}|)}{1 - \alpha}. \end{aligned} \quad (6)$$

Proof. The definition of v^* gives the first inequality, while $v^* \leq \sup_{\bar{\pi}} J(\bar{\pi}, \nu)$ gives the second. Write $A_{\alpha} = \log |\mathcal{A}| / (1 - \alpha)$ and $B_{\alpha} = \log |\mathcal{B}| / (1 - \alpha)$. The per-state entropy bound gives $0 \leq H_{\alpha}^{\pi} \leq A_{\alpha}$ and $0 \leq H_{\alpha}^{\nu} \leq B_{\alpha}$ for every policy pair, irrespective of the induced state distribution. Consequently, for fixed ν ,

$$J(\bar{\pi}, \nu) = J_{\varepsilon}(\bar{\pi}, \nu) - \varepsilon H_{\alpha}^{\pi}(\bar{\pi}, \nu) + \varepsilon H_{\alpha}^{\nu}(\bar{\pi}, \nu) \leq J_{\varepsilon}(\bar{\pi}, \nu) + \varepsilon B_{\alpha},$$

and hence $\sup_{\bar{\pi}} J(\bar{\pi}, \nu) \leq \sup_{\bar{\pi}} J_{\varepsilon}(\bar{\pi}, \nu) + \varepsilon B_{\alpha}$. Likewise $J_{\varepsilon}(\pi, \bar{\nu}) \leq J(\pi, \bar{\nu}) + \varepsilon A_{\alpha}$; consequently, $-\inf_{\bar{\nu}} J(\pi, \bar{\nu}) \leq -\inf_{\bar{\nu}} J_{\varepsilon}(\pi, \bar{\nu}) + \varepsilon A_{\alpha}$. Adding the last two inequalities gives the final bound in (6). $\square$

Thus a global entropy-regularized Nash gap controls the maximizing player’s unregularized worst-case return up to a tunable entropy bias. The stationarity analysis below instead studies the local parameterized field.

Proposition 2 (Smooth finite-game performance component). *Suppose $\mathcal{S}, \mathcal{A}, \mathcal{B}$ are finite, the one-stage rewards are bounded, $\varepsilon > 0$, and, on the parameter neighborhood under consideration, the current policies are C^3 with action probabilities bounded away from zero. Then the entropy-regularized best-response values against either current player, and hence Gap_{ε} , are C^3 functions of the current policy parameters.*

Proof. Fix the minimizing-player policy. The maximizing player’s soft Bellman value satisfies $v = T_{\varepsilon}(v; \psi)$, where each component of T_{ε} is an $\varepsilon \log \sum \exp$ of the action values induced by ν_{ψ} . Explicitly,

$$\begin{aligned} [T_{\varepsilon}(v; \psi)]_s &= \varepsilon \log \sum_a \exp \left(\frac{Q_{\psi}^v(s, a)}{\varepsilon} \right), \\ Q_{\psi}^v(s, a) &\triangleq \bar{r}_{\psi}(s, a) + \alpha \sum_{s'} \bar{P}_{\psi}(s' | s, a) v_{s'}. \end{aligned}$$

Here $\mathcal{H}(p) \triangleq -\sum_i p_i \log p_i$ is the per-state Shannon entropy, $\bar{P}_{\psi}$ averages P over ν_{ψ} , and $\bar{r}_{\psi}$ averages r while including the opponent-entropy term $-\varepsilon \mathcal{H}(\nu_{\psi}(\cdot | s))$ from (5). The map is C^3 in (v, ψ) , and its Jacobian with respect to v is $\alpha P_{\pi_{\text{BR}}, \nu_{\psi}}$, a

stochastic matrix scaled by α . Thus $\|D_v T_\varepsilon\|_\infty \leq \alpha < 1$; therefore, the identity shift of $D_v T_\varepsilon$ is invertible by the Neumann series. The implicit-function theorem makes the unique soft fixed point C^3 in ψ . The minimizing player's soft-min Bellman equation has the same argument after replacing $\log \sum \exp$ by $-\log \sum \exp(-\cdot)$. Multiplication by ρ gives the two regularized best-response values; their difference is Gap_ε and is therefore C^3 . $\square$

In large finite or continuous state spaces, exact dynamic programming is computationally prohibitive or requires solving a functional fixed-point problem. Modern methods therefore replace the policy-space problem by a parameterized one. The maximizing-player policy is represented by $\pi_\theta(a | s)$ with $\theta \in \mathbb{R}^{d_\theta}$, and the minimizing-player policy by $\nu_\psi(b | s)$ with $\psi \in \mathbb{R}^{d_\psi}$, giving

$$J(\theta, \psi) \triangleq \mathbb{E}_{\rho, \pi_\theta, \nu_\psi} [\sum_{t=0}^{\infty} \alpha^t r(s_t, a_t, b_t)], \\ \max_{\theta} \min_{\psi} J(\theta, \psi).$$

In model-free training, stochastic policy-gradient estimators replace the exact gradients using trajectory-derived return, value, or advantage information [25]. For the policy parameters above, $z \in \mathbb{R}^d$ in (1), where $d \triangleq d_\theta + d_\psi$. A point $z^* \in \mathcal{Z}^*$ is max-player ascent-stationary and min-player descent-stationary. Thus $\mathcal{Z}^*$ is the first-order KKT set of the unconstrained parameterized objective. Corollary 3 connects the decrease in $\mathcal{V}$ to the policy-space Nash gap and worst-case return.

The field F in (1) is the *population target field* defined through exact policy gradients. Section II-D models the implemented stochastic oracles and their estimation errors.

We impose the following local regularity, used for the template expansion and the analysis of the one-step change in $\mathcal{V}$.

Assumption 1 (Local smoothness). *There exists an open set $\mathcal{D} \subset \mathbb{R}^d$ such that:*

- (a) *all iterates and probing points used in the analysis lie in $\mathcal{D}$;*
- (b) *$J : \mathbb{R}^d \rightarrow \mathbb{R}$ is C^4 on $\mathcal{D}$;*
- (c) *all derivatives of J up to order four are bounded on $\mathcal{D}$.*

Conditions (b)–(c) imply $F \in C^3(\mathcal{D}; \mathbb{R}^d)$, the regularity used throughout; condition (a) is an admissibility requirement enforced in implementations by the safeguards of Section V.

B. Classical templates and the curvature-induced direction

For a smooth saddle field, the goal is to find z^* with $F(z^*) = 0$. Eq. (2) defines the GDA, PPM, and EGM updates used below. Proposition 3 makes their local second-order relation precise.

The stabilizing mechanism is visible on the bilinear max-min saddle $L(x, y) = -xy$, which is consistent with the sign convention in (1) and has field $F = J_0 z$, where

$$J_0 \triangleq \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

This matrix satisfies $J_0^2 = -I_2$, where I_n denotes the $n \times n$ identity. GDA has eigenvalue modulus $\sqrt{1 + s^2} > 1$ and spirals

outward; PPM contracts for every $s > 0$, while EGM contracts for $0 < s < 1$ (is neutral at $s = 1$ and expansive for $s > 1$). Recall ϕ and G from (3). Below, $\|\cdot\|$ also denotes the induced matrix norm, $\langle \cdot, \cdot \rangle$ denotes the standard inner product, and $\nabla^2 F$ denotes the second-derivative tensor.

Proposition 3 (Discrete-time second-order template expansions). *Suppose Assumption 1 holds and $s > 0$ is small enough that the PPM and EGM updates are well defined. Then*

$$\text{GDA: } z_+ - z = -sF(z),$$

$$\text{PPM: } z_+ - z = -sF(z) + s^2 G(z) + R_s^{\text{PPM}}(z),$$

$$\text{EGM: } z_+ - z = -sF(z) + s^2 G(z) + R_s^{\text{EGM}}(z),$$

with $\|R_s^{\text{PPM}}(z)\| \leq C_{\text{PPM}} s^3$ and $\|R_s^{\text{EGM}}(z)\| \leq C_{\text{EGM}} s^3$ for finite constants depending only on local bounds of F , ∇F , and $\nabla^2 F$ on $\mathcal{D}$.

Proof. See Appendix A. $\square$

PPM and EGM both contain the correction $+s^2 G(z)$, which helps explain their local stability advantage over GDA in the bilinear example. On the bilinear example $G = -\nabla \phi$; therefore, $+G$ is the steepest-descent direction of ϕ , whereas $-F$ is tangent to its level sets. The expansion couples the two directions through a single s ; the proposed method retains both but decouples their magnitudes.

C. When and why the curvature direction helps

Fix $z \in \mathcal{D}$, write $A \triangleq \nabla F(z)$ for the local Jacobian, and split it into its symmetric and antisymmetric parts

$$A = S + W,$$

$$S \triangleq \text{sym}(A) \triangleq \frac{1}{2}(A + A^\top),$$

$$W \triangleq \text{skew}(A) \triangleq \frac{1}{2}(A - A^\top).$$

In differentiable-games terminology, S is the potential-like component and W the Hamiltonian component [17]. Lemma 1 expresses the one-step field-energy directional derivatives exactly in terms of S and W .

Lemma 1 (Symmetric–antisymmetric decomposition of the field-energy derivatives). *Let $\phi(z) = \frac{1}{2} \|F(z)\|^2$; then $\nabla \phi(z) = A^\top F(z)$ with $A = \nabla F(z)$. Then the directional derivatives of ϕ along $-F$ and along $+G = AF$ satisfy*

$$\langle \nabla \phi(z), -F(z) \rangle = -F(z)^\top S F(z), \quad (7)$$

$$\langle \nabla \phi(z), G(z) \rangle = \|S F(z)\|^2 - \|W F(z)\|^2. \quad (8)$$

Proof. See Appendix A. $\square$

The compact form of (8) uses the cancellation $F^\top (SW + WS) F = 0$, since $SW + WS$ is antisymmetric (Appendix A). Eq. (7) shows that the field direction decreases field energy through the symmetric part: in a purely rotational field ($S = 0$), $-F$ is tangent to the level sets of ϕ , exactly the GDA cycling mechanism. Eq. (8) balances the symmetric action $\|S F\|^2$ against the antisymmetric action $\|W F\|^2$, yielding the following condition.

Theorem 1 (Skew-dominance characterization of curvature descent). *Fix $z \in \mathcal{D}$ with $F(z) \neq 0$, and let $A = S + W$*

as above. The curvature direction $+\mathbf{G}(\mathbf{z})$ is a strict descent direction for the field energy ϕ at $\mathbf{z}$, i.e. $\langle \nabla \phi(\mathbf{z}), \mathbf{G}(\mathbf{z}) \rangle < 0$, if and only if the antisymmetric action dominates the symmetric action along $\mathbf{F}$:

$$\|\mathbf{WF}(\mathbf{z})\| > \|\mathbf{SF}(\mathbf{z})\|. \quad (9)$$

In particular:

- (a) if the field is purely rotational at $\mathbf{z}$ ($\mathbf{SF}(\mathbf{z}) = \mathbf{0}$ and $\mathbf{WF}(\mathbf{z}) \neq \mathbf{0}$), then $\langle \nabla \phi, -\mathbf{F} \rangle = 0$ and $\langle \nabla \phi, \mathbf{G} \rangle = -\|\mathbf{WF}(\mathbf{z})\|^2 < 0$; hence $+\mathbf{G}$ strictly decreases ϕ while $-\mathbf{F}$ does not;
- (b) if the field is symmetric at $\mathbf{z}$ ($\mathbf{W} = \mathbf{0}$), then $\langle \nabla \phi, \mathbf{G} \rangle = \|\mathbf{SF}(\mathbf{z})\|^2 \geq 0$; hence $+\mathbf{G}$ is not a descent direction for ϕ and acts as an ordinary curvature term;
- (c) if $\mathbf{A}$ is normal and has a two-dimensional real invariant subspace with an orthonormal basis in which its restriction is $\mu\mathbf{I} + \sigma J_0$, with $\mathbf{F}(\mathbf{z})$ in that subspace, then (9) reduces to $|\sigma| > |\mu|$, i.e. the rotational rate exceeds the contraction rate.

Proof. See Appendix A. $\square$

Theorem 1 gives an exact local criterion. The curvature direction decreases field energy when the antisymmetric Jacobian action along $\mathbf{F}$ is larger than the symmetric action. For the normal two-dimensional field, this condition reduces to $|\sigma| > |\mu|$. Eq. (7) also explains why the field direction alone cannot decrease field energy in a purely rotational field. Jacobian–vector and transposed-Jacobian–vector products evaluate this condition and yield the diagnostics $\|\mathbf{WF}\|$ and $\|\mathbf{SF}\|$ used in Section VI.

Remark 1 (Field-energy geometry and the composite Lyapunov function). Theorem 1 studies the intrinsic geometry of $+\mathbf{G}$ using the Euclidean field energy ϕ . For $\mathcal{V} = \phi$, skew dominance is necessary and sufficient for $+\mathbf{G}$ to be a strict descent direction. For a general composite function $\mathcal{V}$, the first-order change along $+\mathbf{G}$ is $\langle \nabla \mathcal{V}, \mathbf{G} \rangle$, and the QP measures the additional decrease in its local model relative to the field-only restriction. A nonidentity block weight also changes the Euclidean decomposition. Theorems 1, 3, and 6 therefore connect field-energy geometry, local one-step decrease in $\mathcal{V}$, and the finite-time stationarity bound.

Corollary 1 connects Theorem 1 to the worst-case bilinear picture and the collinearity diagnostic.

Corollary 1 (Local linear field and collinearity). *Let $\mathbf{F}(\mathbf{z}) = \mathbf{A}(\mathbf{z} - \mathbf{z}^*)$ near $\mathbf{z}^*$, with $\mathbf{A} = \mathbf{S} + \mathbf{W}$, and fix $\mathbf{z}$ with $\mathbf{F}(\mathbf{z}) \neq \mathbf{0}$. Then $\mathbf{G}(\mathbf{z}) = \mathbf{A}\mathbf{F}(\mathbf{z})$, and $\mathbf{G}(\mathbf{z})$ is collinear with $\mathbf{F}(\mathbf{z})$, i.e. $\mathbf{G}(\mathbf{z}) = \lambda\mathbf{F}(\mathbf{z})$ for some real λ , if and only if $\mathbf{F}(\mathbf{z})$ is a real eigenvector of $\mathbf{A}$. If $\mathbf{A}$ is skew-symmetric and $\mathbf{A}^2 = -\sigma^2\mathbf{I}$ on the invariant subspace containing $\mathbf{z} - \mathbf{z}^*$, then $\mathbf{F}(\mathbf{z})$ and $\mathbf{G}(\mathbf{z})$ are orthogonal, $\langle \nabla \phi, -\mathbf{F} \rangle = 0$, and $\langle \nabla \phi, \mathbf{G} \rangle = -\sigma^4 \|\mathbf{z} - \mathbf{z}^*\|^2$.*

Proof. See Appendix A. $\square$

Corollary 1 isolates two regimes. When the current field is dominated by one real eigendirection of $\mathbf{A}$, $\mathbf{G}$ is nearly collinear with $\mathbf{F}$; for a purely antisymmetric Jacobian, $\mathbf{F}^\top \mathbf{G} = \mathbf{F}^\top \mathbf{A}\mathbf{F} = 0$ for every $\mathbf{F}$. Section III-C shows that, when

the pure-field coordinate is itself descending, strict predicted-decrease advantage is equivalent to a nonzero curvature reduced gradient in the local Lyapunov metric. Non-collinearity indicates that the two directions enlarge the local search span, but it does not guarantee a QP gain. The gain also depends on the Lyapunov gradient and the local quadratic model. Thus the angle between $\mathbf{F}$ and $\mathbf{G}$ is a useful diagnostic, but it does not by itself determine whether the curvature coordinate is beneficial.

Fig. 1 isolates the field geometry. Section VI tests the predicted skew threshold, the two-coordinate QP identity, and the performance consequences.

D. Adaptive stochastic recursion and equilibrium notion

Let $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$ denote the stochastic field and curvature oracles at update k , respectively. We study the recursion

$$\mathbf{z}_{k+1} = \mathbf{z}_k - \beta_k \hat{\mathbf{F}}_k + \gamma_k \hat{\mathbf{G}}_k, \quad (10)$$

where $\beta_k, \gamma_k \geq 0$ are adaptive step sizes selected by minimizing a local model of the one-step change in $\mathcal{V}$. Let $\mathcal{F}_k$ denote the filtration generated by all algorithmic history before the stochastic queries at iteration k .

Assumption 2 (Biased stochastic oracles). *For every k ,*

$$\hat{\mathbf{F}}_k = \mathbf{F}(\mathbf{z}_k) + \mathbf{b}_k^F + \boldsymbol{\xi}_k^F, \quad \hat{\mathbf{G}}_k = \mathbf{G}(\mathbf{z}_k) + \mathbf{b}_k^G + \boldsymbol{\xi}_k^G,$$

where $\mathbf{b}_k^F, \mathbf{b}_k^G$ are $\mathcal{F}_k$ -measurable and $\mathbb{E}[\boldsymbol{\xi}_k^F | \mathcal{F}_k] = \mathbb{E}[\boldsymbol{\xi}_k^G | \mathcal{F}_k] = \mathbf{0}$. There exist finite $\mathcal{F}_k$ -measurable scalars $\bar{b}_k^F, \bar{b}_k^G, \sigma_{2,k}^F, \sigma_{3,k}^F \geq 0$ such that, almost surely,

- (a) $\|\mathbf{b}_k^F\| \leq \bar{b}_k^F$ and $\|\mathbf{b}_k^G\| \leq \bar{b}_k^G$;
- (b) $\mathbb{E}[\|\boldsymbol{\xi}_k^F\|^2 + \|\boldsymbol{\xi}_k^G\|^2 | \mathcal{F}_k] \leq \sigma_{2,k}^2$;
- (c) $\mathbb{E}[\|\hat{\mathbf{F}}_k\|^3 + \|\hat{\mathbf{G}}_k\|^3 | \mathcal{F}_k] \leq \sigma_{3,k}^3$.

Assumption 2 covers unbiased mini-batch estimators and biases from clipped surrogates, truncated rollouts, and finite-difference curvature oracles. Section V treats directional-secant and moment-smoothed estimates of this local model through the estimation-error bound in Proposition 6. The convergence target is the population stationarity set $\mathcal{Z}^*$. Section III defines the weighted stationarity residual used in the finite-time analysis.

III. ADAPTIVE STEP-SIZE DESIGN FROM THE ONE-STEP CHANGE IN A COMPOSITE LYAPUNOV FUNCTION

A. Composite Lyapunov function and second-order expansion

Here, Lyapunov drift means the one-step change $\mathcal{V}(\mathbf{z}_{k+1}) - \mathcal{V}(\mathbf{z}_k)$ in the function defined below.

A small field norm need not imply a small policy-space gap. We therefore combine field energy with a regularized local saddle-gap term and an optional nonnegative task-specific error.

Definition 1 (Local proximal saddle-gap residual). *Fix proximal parameters $\tau_\theta, \tau_\psi > 0$. For every $\mathbf{z}$, let $\mathcal{B}_\theta(\mathbf{z})$ and $\mathcal{B}_\psi(\mathbf{z})$ be nonempty compact local neighborhoods containing $\boldsymbol{\theta}$ and $\boldsymbol{\psi}$, respectively. For $\mathbf{z} = (\boldsymbol{\theta}, \boldsymbol{\psi})$, define the maximizing and minimizing players' local proximal improvement gaps*

$$\mathcal{G}_{\theta, \tau}(\mathbf{z}) \triangleq \max_{\bar{\boldsymbol{\theta}} \in \mathcal{B}_\theta(\mathbf{z})} \left\{ J(\bar{\boldsymbol{\theta}}, \boldsymbol{\psi}) - \frac{1}{2\tau_\theta} \|\bar{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 \right\} - J(\boldsymbol{\theta}, \boldsymbol{\psi}),$$

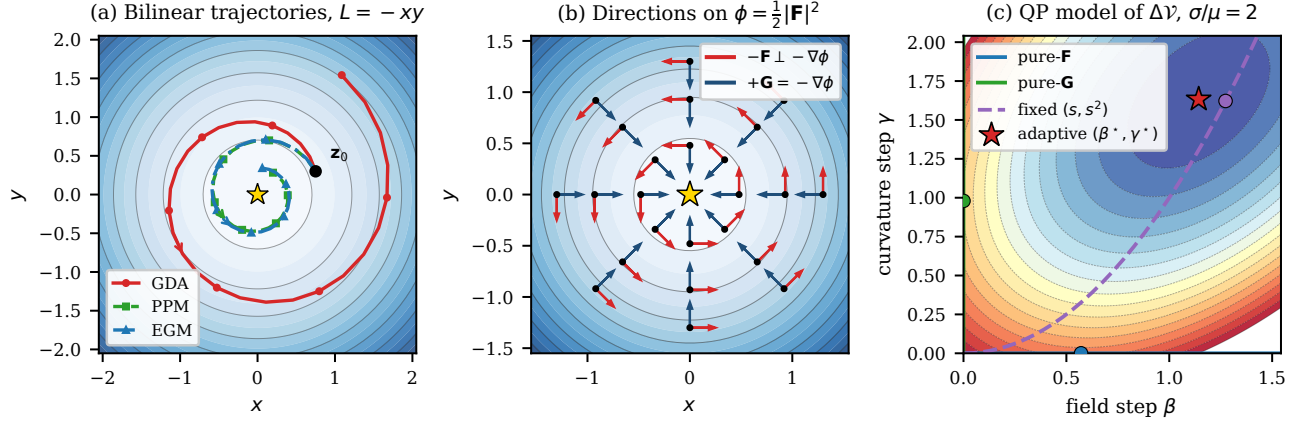


Fig. 1. Analytical mechanism illustration from exact calculations. (a) On the bilinear game $L = -xy$, GDA spirals outward, whereas PPM and EGM contract toward the saddle ($s = 0.25$). (b) For $\mathbf{F}(x, y) = (y, -x)$, $-\mathbf{F}$ is tangent to the field-energy level sets, while $+\mathbf{G} = \nabla\mathbf{F}\mathbf{F} = -\nabla\phi$ points inward. (c) The QP searches the nonnegative (β, γ) plane rather than a coordinate ray or the fixed EGM coupling (s, s^2) ; contours use $\mathcal{V} = \phi$, $\mu = 0.35$, $\sigma = 0.70$, and $\mathbf{z} = (1, -0.65)$ without step size caps.

$$\mathcal{G}_{\psi, \tau}(\mathbf{z}) \triangleq J(\boldsymbol{\theta}, \boldsymbol{\psi}) - \min_{\bar{\boldsymbol{\psi}} \in B_{\psi}(\mathbf{z})} \left\{ J(\boldsymbol{\theta}, \bar{\boldsymbol{\psi}}) + \frac{1}{2\tau_{\psi}} \|\bar{\boldsymbol{\psi}} - \boldsymbol{\psi}\|^2 \right\},$$

The compactness assumptions make the extrema well defined, and inclusion of the current parameters gives $\mathcal{G}_{\psi, \tau}(\mathbf{z}) \geq 0$ and $\mathcal{G}_{\psi, \tau}(\mathbf{z}) \geq 0$. Fix a smoothing parameter $\epsilon > 0$. The regularized local saddle-gap residual is

$$\mathcal{P}_{\tau}(\mathbf{z}) \triangleq \tilde{\sigma}_{\epsilon}(\mathcal{G}_{\psi, \tau}(\mathbf{z})) + \tilde{\sigma}_{\epsilon}(\mathcal{G}_{\psi, \tau}(\mathbf{z})),$$

where

$$\tilde{\sigma}_{\epsilon}(u) \triangleq \epsilon \log \left(\frac{1 + e^{u/\epsilon}}{2} \right), \quad u \geq 0,$$

is the centered softplus. It is C^{∞} , nonnegative on the nonnegative gap domain, $\tilde{\sigma}_{\epsilon}(0) = 0$, and $0 \leq u - \tilde{\sigma}_{\epsilon}(u) \leq \epsilon \log 2$ for $u \geq 0$.

Let $\mathcal{P}$ denote the nonnegative C^3 performance component used by the controller. We take $\mathcal{P} = \mathcal{P}_{\tau}$ for the local realization above and $\mathcal{P} = \text{Gap}_{\epsilon}$ when global regularized best responses are available. The latter bounds the maximizing player's worst-case-return suboptimality through (6).

Remark 2 (Smooth realization of $\mathcal{P}_{\tau}$). The analysis requires the chosen function $\mathcal{V}$ to be C^3 . For exact envelopes, this follows locally when the feasible-set parameterizations are C^3 , the optimizers are unique and interior, and their second-order optimality matrices are nonsingular. A fixed differentiable inner loop also suffices when nonnegativity is enforced by construction. Hard safeguards require a fixed active set on the analyzed neighborhood.

Definition 2 (Composite Lyapunov function). Let $\eta_{\psi} > 0$ and define the block-diagonal weight matrix

$$\mathbf{W}_V \triangleq \text{diag}(\mathbf{I}_{d_{\theta}}, \eta_{\psi} \mathbf{I}_{d_{\psi}}) \in \mathbb{R}^{d \times d},$$

where $\text{diag}(\cdot)$ forms a block-diagonal matrix. The weighted field-energy component is $\mathcal{E}_F(\mathbf{z}) \triangleq \frac{1}{2} \mathbf{F}(\mathbf{z})^{\top} \mathbf{W}_V \mathbf{F}(\mathbf{z})$. Let $\mathcal{C}(\mathbf{z})$ be any nonnegative C^3 task-specific scalar error. For fixed normalizers $\kappa_F, \kappa_P, \kappa_C > 0$ and weights $\lambda_F > 0$, $\lambda_P \geq 0$, $\lambda_C \geq 0$, define the composite Lyapunov function

$$\mathcal{V}(\mathbf{z}) \triangleq \lambda_F \frac{\mathcal{E}_F(\mathbf{z})}{\kappa_F} + \lambda_P \frac{\mathcal{P}(\mathbf{z})}{\kappa_P} + \lambda_C \frac{\mathcal{C}(\mathbf{z})}{\kappa_C}. \quad (11)$$

The experiments use a fixed numerical floor $\epsilon_0 > 0$ and initial-point normalizers, e.g., $\kappa_F = \mathcal{E}_F(\mathbf{z}_0) + \epsilon_0$, $\kappa_P = \mathcal{P}(\mathbf{z}_0) + \epsilon_0$, and $\kappa_C = \mathcal{C}(\mathbf{z}_0) + \epsilon_0$; they set $\lambda_C = 0$.

Here $\mathbf{W}_V$ weights the two policy-gradient blocks and is unrelated to the antisymmetric Jacobian part $\mathbf{W} = \text{skew}(\nabla \mathbf{F})$. All experiments use $\eta_{\psi} = 1$, which gives $\mathcal{E}_F = \phi$; a general η_{ψ} permits fixed block scaling. Setting $\lambda_P = \lambda_C = 0$ gives the field-energy-only special case used in Sections VI-A and VI-D. The field-energy term is always present because $\lambda_F > 0$. The other terms add performance or task-specific information. The conditions below ensure that the expected one-step change in $\mathcal{V}$ is negative up to stochastic and approximation errors.

Lemma 2 (Properties of the composite Lyapunov function). Suppose Assumption 1 holds and $\mathcal{P}, \mathcal{C}$ are C^3 on $\mathcal{D}$. Then:

- (a) $\mathcal{V}(\mathbf{z}) \geq 0$ on $\mathcal{D}$, and since $\lambda_F > 0$, $\mathcal{V}(\mathbf{z}) = 0$ implies $\mathbf{z} \in \mathcal{Z}^*$;
- (b) $\mathcal{V} \in C^3(\mathcal{D})$;
- (c) the gradient is $\nabla \mathcal{V} = (\lambda_F/\kappa_F) \nabla \mathcal{E}_F + (\lambda_P/\kappa_P) \nabla \mathcal{P} + (\lambda_C/\kappa_C) \nabla \mathcal{C}$, where $\nabla \mathcal{E}_F = \nabla \mathbf{F}^{\top} \mathbf{W}_V \mathbf{F}$;
- (d) $\Phi(\mathbf{z}) \triangleq \mathbf{F}(\mathbf{z})^{\top} \mathbf{W}_V \mathbf{F}(\mathbf{z})$ is the weighted stationarity residual and satisfies $\Phi(\mathbf{z}) \leq 2\kappa_F \mathcal{V}(\mathbf{z})/\lambda_F$.

Proof. See Appendix A. $\square$

The implication $\mathcal{V}(\mathbf{z}) = 0 \Rightarrow \mathbf{F}(\mathbf{z}) = 0$ always holds. A positive gap term can keep $\mathcal{V}(\mathbf{z}) > 0$ at a stationary point. We use $\mathcal{V}$ to control the one-step update and Φ to measure first-order stationarity.

For nonnegative (β, γ) , the trial displacement is $\Delta_k(\beta, \gamma) \triangleq -\beta \hat{\mathbf{F}}_k + \gamma \hat{\mathbf{G}}_k$.

Assumption 3 (Local regularity along the iterates). For every k and every feasible (β, γ) produced by the step-size routine, the segment $\{\mathbf{z}_k + t\Delta_k(\beta, \gamma) : t \in [0, 1]\} \subset \mathcal{D}$. There exist $\mathcal{F}_k$ -measurable $G_{V,k}, H_{V,k}, M_{3,k} \geq 0$ that, simultaneously for every feasible pair and almost every realization of the current stochastic queries, satisfy $\|\nabla \mathcal{V}(\mathbf{z}_k)\| \leq G_{V,k}$, $\|\nabla^2 \mathcal{V}(\mathbf{z}_k)\| \leq H_{V,k}$, and $\sup_{t \in [0, 1]} \|\nabla^3 \mathcal{V}(\mathbf{z}_k + t\Delta_k(\beta, \gamma))\| \leq M_{3,k}$.

The five local-model terms are

$$\begin{aligned} e_k &\triangleq \langle \nabla \mathcal{V}(\mathbf{z}_k), \widehat{\mathbf{F}}_k \rangle, & d_k &\triangleq -\langle \nabla \mathcal{V}(\mathbf{z}_k), \widehat{\mathbf{G}}_k \rangle, \\ c_k &\triangleq \widehat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \widehat{\mathbf{F}}_k, & a_k &\triangleq \widehat{\mathbf{G}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \widehat{\mathbf{G}}_k, \\ b_k &\triangleq \widehat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \widehat{\mathbf{G}}_k. \end{aligned} \quad (12)$$

These terms are directional derivatives of $\mathcal{V}$ in (11). Thus, the quadratic model accounts for every selected component of $\mathcal{V}$ rather than only field energy.

Lemma 3 (Second-order expansion of the one-step change in $\mathcal{V}$). *Under Assumption 3, for every feasible (β, γ) ,*

$$\begin{aligned} &\mathcal{V}(\mathbf{z}_k + \Delta_k(\beta, \gamma)) - \mathcal{V}(\mathbf{z}_k) \\ &= -\beta e_k - \gamma d_k + \frac{1}{2} c_k \beta^2 - b_k \beta \gamma + \frac{1}{2} a_k \gamma^2 + \text{rem}_k(\beta, \gamma), \end{aligned} \quad (13)$$

$$\text{with } |\text{rem}_k(\beta, \gamma)| \leq \frac{M_{3,k}}{6} (\beta \|\widehat{\mathbf{F}}_k\| + \gamma \|\widehat{\mathbf{G}}_k\|)^3.$$

Proof. See Appendix A. $\square$

The five local terms in this lemma define the QP below.

B. Implemented quadratic subproblem and closed-form solution

Let $(\hat{e}_k, \hat{d}_k, \hat{c}_k, \hat{a}_k, \hat{b}_k)$ denote the stabilized local-model terms constructed below. With nonnegative diagonal stabilizers $\rho_{a,k}, \rho_{c,k} \geq 0$, the implemented quadratic model of the change in $\mathcal{V}$ is

$$\begin{aligned} \hat{q}_k(\beta, \gamma) &\triangleq -\beta \hat{e}_k - \gamma \hat{d}_k + \frac{1}{2} \hat{c}_k \beta^2 - \hat{b}_k \beta \gamma \\ &\quad + \frac{1}{2} \hat{a}_k \gamma^2, \quad (\beta, \gamma) \in \mathbb{R}_+^2, \end{aligned} \quad (14)$$

where Section V supplies raw estimates $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k)$ of $(e_k, d_k, c_k, a_k, b_k)$. For a scalar u , let $[u]_+ \triangleq \max\{u, 0\}$. For a fixed $\lambda_{\text{pd}} > 0$, define

$$\widetilde{\mathbf{H}}_k = \begin{bmatrix} \tilde{c}_k & -\tilde{b}_k \\ -\tilde{b}_k & \tilde{a}_k \end{bmatrix}.$$

We set

$$\lambda_k = [\lambda_{\text{pd}} - \lambda_{\min}(\widetilde{\mathbf{H}}_k)]_+, \quad \widehat{\mathbf{H}}_k = \widetilde{\mathbf{H}}_k + \lambda_k \mathbf{I}_2.$$

We then set $\hat{e}_k = \tilde{e}_k$, $\hat{d}_k = \tilde{d}_k$, $\hat{b}_k = \tilde{b}_k$, $\hat{c}_k = \tilde{c}_k + \lambda_k$, $\hat{a}_k = \tilde{a}_k + \lambda_k$, and $\rho_{c,k} = \rho_{a,k} = \lambda_k$. Then $\hat{\delta}_k \triangleq \det \widehat{\mathbf{H}}_k = \hat{c}_k \hat{a}_k - \hat{b}_k^2$, and (β_k^*, γ_k^*) minimizes $\hat{q}_k$ over $\mathbb{R}_+^2$.

Assumption 4 (Positive definiteness). *For every k , $\widehat{\mathbf{H}}_k \succ 0$, equivalently $\hat{c}_k > 0$ and $\hat{\delta}_k > 0$.*

The common spectral shift above gives $\widehat{\mathbf{H}}_k \succeq \lambda_{\text{pd}} \mathbf{I}_2$ and therefore enforces Assumption 4. The unconstrained critical point is

$$(\beta_k^{\text{uc}}, \gamma_k^{\text{uc}}) \triangleq \left(\frac{\hat{a}_k \hat{e}_k + \hat{b}_k \hat{d}_k}{\hat{\delta}_k}, \frac{\hat{b}_k \hat{e}_k + \hat{c}_k \hat{d}_k}{\hat{\delta}_k} \right), \quad (15)$$

and the boundary candidates are $([\hat{e}_k/\hat{c}_k]_+, 0)$, $(0, [\hat{d}_k/\hat{a}_k]_+)$, $(0, 0)$.

Theorem 2 (Closed-form implemented step-size rule). *Under Assumption 4, the implemented problem (14) admits a unique minimizer, given by (15) when both components are nonnegative, and otherwise by the boundary candidate with smallest $\hat{q}_k$.*

When the unconstrained critical point belongs to $\mathbb{R}_+^2$, the implemented predicted decrease $\widehat{\Delta}_k^* \triangleq -\hat{q}_k(\beta_k^*, \gamma_k^*)$ is

$$\widehat{\Delta}_k^* = \frac{\hat{a}_k \hat{e}_k^2 + 2\hat{b}_k \hat{e}_k \hat{d}_k + \hat{c}_k \hat{d}_k^2}{2\hat{\delta}_k}. \quad (16)$$

Proof. See Appendix A. $\square$

Because the subproblem is two-dimensional, the unconstrained formula and boundary fallback use a fixed number of scalar operations. The rule depends only on the current local model estimates.

Proposition 4 (Exact box-constrained fallback). *Under Assumption 4, fix finite caps $\beta_{\max}, \gamma_{\max} > 0$. The unique minimizer of $\hat{q}_k$ on $[0, \beta_{\max}] \times [0, \gamma_{\max}]$ is the member with smallest objective value among the feasible unconstrained point (15) and the four edge minimizers*

$$\begin{aligned} &([\hat{e}_k/\hat{c}_k]_{[0, \beta_{\max}]}, 0), \quad (0, [\hat{d}_k/\hat{a}_k]_{[0, \gamma_{\max}]}), \\ &(\beta_{\max}, [\hat{d}_k + \hat{b}_k \beta_{\max}]/\hat{a}_k]_{[0, \gamma_{\max}]}), \\ &([\hat{e}_k + \hat{b}_k \gamma_{\max}]/\hat{c}_k]_{[0, \beta_{\max}]}, \gamma_{\max}), \end{aligned}$$

where $[u]_{[l, r]} \triangleq \min\{r, \max\{l, u\}\}$. Consequently its predicted decrease dominates every one-direction or fixed-coupling comparator whose step size pair lies in the same box.

Proof. Strict convexity gives a unique minimizer. If it is in the interior, the gradient equations give (15); otherwise it lies on one of the four edges, and differentiating the resulting one-dimensional strictly convex quadratic and clipping its stationary point to that edge gives the four displayed candidates. Set inclusion then proves the dominance statement. $\square$

C. Predicted-decrease dominance and its geometric meaning

Theorem 3 (Predicted-decrease dominance). *Under Assumption 4, fix $s_{\max} > 0$ and define*

$$\widehat{\Delta}_k^{\mathbf{F}} \triangleq -\min_{\beta \geq 0} \hat{q}_k(\beta, 0) = \frac{[\hat{e}_k]_+^2}{2\hat{c}_k},$$

$$\widehat{\Delta}_k^{\mathbf{G}} \triangleq -\min_{\gamma \geq 0} \hat{q}_k(0, \gamma) = \frac{[\hat{d}_k]_+^2}{2\hat{a}_k},$$

$$\widehat{\Delta}_k^{2\text{nd}} \triangleq -\min_{s \in [0, s_{\max}]} \hat{q}_k(s, s^2),$$

Then $\widehat{\Delta}_k^* \geq \max\{\widehat{\Delta}_k^{\mathbf{F}}, \widehat{\Delta}_k^{\mathbf{G}}, \widehat{\Delta}_k^{2\text{nd}}\}$ almost surely. Moreover, when the unconstrained critical point (15) belongs to $\mathbb{R}_+^2$, the gap over the pure-field rule is exactly

$$\widehat{\Delta}_k^* - \widehat{\Delta}_k^{\mathbf{F}} = \frac{(\hat{b}_k \hat{e}_k + \hat{c}_k \hat{d}_k)^2}{2\hat{c}_k \hat{\delta}_k} + \frac{[-\hat{e}_k]_+^2}{2\hat{c}_k} \geq 0, \quad (17)$$

which is strictly positive if and only if either $\hat{b}_k \hat{e}_k + \hat{c}_k \hat{d}_k \neq 0$ or $\hat{e}_k < 0$. In the usual field-descent regime $\hat{e}_k \geq 0$, strict improvement is therefore equivalent to the curvature direction having a nonzero reduced gradient after projecting out the field direction in the $\widehat{\mathbf{H}}_k$ -metric.

Proof. See Appendix A. $\square$

At an interior solution, (17) quantifies the extra predicted decrease supplied by the curvature coordinate. The reduced gradient $\hat{b}_k \hat{e}_k + \hat{c}_k \hat{d}_k$ is the component that remains after accounting for the field direction in the local quadratic model. If $\mathbf{G}$ is collinear with $\mathbf{F}$, the two directions span one geometric direction. With zero diagonal inflation, the associated two-coordinate Hessian is singular. Any QP gain in this case comes from step size constraints or regularization, not span enlargement. In the field-descent regime $\hat{e}_k \geq 0$, antisymmetry alone is insufficient: strict interior improvement also requires a nonzero reduced descent component. Assumption 7 measures this improvement in the stochastic stationarity bound.

D. Pathwise one-step bound and per-iteration error

The implemented model is allowed to differ from the reference quadratic

$$q_k(\beta, \gamma) \triangleq -\beta e_k - \gamma d_k + \frac{1}{2}(c_k + \rho_{c,k})\beta^2 - b_k \beta \gamma + \frac{1}{2}(a_k + \rho_{a,k})\gamma^2$$

through a one-sided model-consistency residual. Define the cubic Taylor remainder $\varepsilon_k^{\text{rem}} \triangleq M_{3,k}(\beta_k \|\hat{\mathbf{F}}_k\| + \gamma_k \|\hat{\mathbf{G}}_k\|)^3/6$. Let $\varepsilon_k^{\text{model}} \geq 0$ be $\mathcal{F}_{k+1}$ -measurable and satisfy $q_k(\beta_k, \gamma_k) \leq \hat{q}_k(\beta_k, \gamma_k) + \varepsilon_k^{\text{model}}$, and set $\varepsilon_k^{\text{step}} \triangleq \hat{q}_k(\beta_k, \gamma_k) + \hat{\Delta}_k^* \geq 0$. The total per-iteration residual is

$$\varepsilon_k^{\text{tot}} \triangleq \varepsilon_k^{\text{rem}} + \varepsilon_k^{\text{model}} + \varepsilon_k^{\text{step}}, \quad (18)$$

The residual $\varepsilon_k^{\text{model}}$ vanishes for exact autodifferentiation and admits an explicit bound for approximate realizations through Proposition 6; $\varepsilon_k^{\text{step}}$ vanishes for the unsafeguarded closed-form rule.

Lemma 4 (Pathwise bound on the one-step change in $\mathcal{V}$). *Under Assumptions 3 and 4, for every k ,*

$$\mathcal{V}(\mathbf{z}_{k+1}) - \mathcal{V}(\mathbf{z}_k) \leq -\hat{\Delta}_k^* + \varepsilon_k^{\text{tot}} \quad \text{almost surely.} \quad (19)$$

Proof. See Appendix A. $\square$

The inequality separates model decrease from implementation residuals.

Lemma 5 (Conditional residual bound). *Suppose Assumptions 2 and 3 hold and $\beta_k \leq \beta_{\max}$, $\gamma_k \leq \gamma_{\max}$. Then*

$$\mathbb{E}[\varepsilon_k^{\text{rem}} \mid \mathcal{F}_k] \leq \frac{2M_{3,k}}{3}(\beta_{\max} + \gamma_{\max})^3 \sigma_{3,k}^3. \quad (20)$$

If additionally $M_{3,k} \leq \bar{M}_3$ and $\sigma_{3,k} \leq \bar{\sigma}_3$, then $\mathbb{E}[\varepsilon_k^{\text{rem}}]$ is uniformly bounded.

Proof. See Appendix A. $\square$

Thus the cubic remainder has a finite conditional expectation under the stated moment bound.

IV. EXPECTED ONE-STEP CHANGE AND CONVERGENCE ANALYSIS

A. Local admissible region and structural conditions

Recall the weighted stationarity residual

$$\Phi(\mathbf{z}) \triangleq \mathbf{F}(\mathbf{z})^\top \mathbf{W}_V \mathbf{F}(\mathbf{z}) = \|\nabla_{\theta} J\|^2 + \eta_{\psi} \|\nabla_{\psi} J\|^2,$$

which vanishes precisely on $\mathcal{Z}^*$.

Assumption 5 (Local admissible region and Lyapunov decrease). *There exist an open admissible set $\mathcal{N}$ whose compact closure satisfies $\bar{\mathcal{N}} \subset \mathcal{D}$, an index $k_0 \geq 0$, and constants $\mu_{\Phi} \geq 0$, $\mu_V \geq 0$ such that $\mathcal{N}$ contains the local trajectory after k_0 together with the stationary component of $\mathcal{Z}^*$ under consideration, and:*

- (a) $\mathbf{z}_k \in \mathcal{N}$ for all $k \geq k_0$ almost surely;
- (b) $\langle \nabla \mathcal{V}(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle \geq \mu_{\Phi} \Phi(\mathbf{z})$ for all $\mathbf{z} \in \mathcal{N}$;
- (c) when geometric contraction is invoked, $\langle \nabla \mathcal{V}(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle \geq \mu_V \mathcal{V}(\mathbf{z})$ for all $\mathbf{z} \in \mathcal{N}$.

Because the algorithm moves along $-\mathbf{F}$, condition (b) states that this direction decreases $\mathcal{V}$ at least in proportion to the stationarity residual Φ . Combined with the conditional bound below on the expected one-step change in $\mathcal{V}$, it gives the finite-time stationarity result. It plays the role of a Polyak–Łojasiewicz (PL) inequality in a gradient system [26]. Condition (a) is a local invariance assumption. A prescribed trust region and backtracking can keep trial points in a chosen compact region. The analysis requires that region to satisfy (b)–(c). Condition (c) strengthens the required decrease from a multiple of Φ to a multiple of $\mathcal{V}$ itself; this stronger condition gives geometric contraction of $\mathbb{E}[\mathcal{V}]$. It requires $\mathcal{V}$ to vanish on the target component, excluding stationary points with a positive local gap from the contraction result.

Lemma 6 (Sufficient condition for Lyapunov decrease). *Suppose Assumption 1 holds and there exist $m_F > 0$ and $\delta_P, \delta_C \geq 0$ such that, for all $\mathbf{z} \in \mathcal{N}$,*

$$\mathbf{F}(\mathbf{z})^\top \text{sym}(\mathbf{W}_V \nabla \mathbf{F}(\mathbf{z})) \mathbf{F}(\mathbf{z}) \geq m_F \mathbf{F}(\mathbf{z})^\top \mathbf{W}_V \mathbf{F}(\mathbf{z}), \quad (21)$$

and

$$\frac{\lambda_P}{\kappa_P} \langle \nabla \mathcal{P}(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle + \frac{\lambda_C}{\kappa_C} \langle \nabla \mathcal{C}(\mathbf{z}), \mathbf{F}(\mathbf{z}) \rangle \geq -(\delta_P + \delta_C) \Phi(\mathbf{z}). \quad (22)$$

If $\lambda_F m_F / \kappa_F > \delta_P + \delta_C$, then Assumption 5(b) holds with

$$\mu_{\Phi} = \lambda_F m_F / \kappa_F - \delta_P - \delta_C > 0.$$

Proof. See Appendix A. $\square$

Remark 3 (Compatibility of the Lyapunov components). Eq. (21) is positivity of the weighted symmetric Jacobian along $\mathbf{F}$; in the canonical unweighted pure-rotation case $\mathbf{W}_V = \mathbf{I}$, one has $m_F = 0$. Bounded gap gradients alone do not imply (22), whose right side is $O(\|\mathbf{F}\|^2)$: the negative gap/auxiliary derivatives must vanish at least linearly with $\mathbf{F}$. The resulting δ_P, δ_C scale with their component weights, making the displayed strict inequality an explicit compatibility condition.

Because $\bar{\mathcal{N}}$ is compactly contained in $\mathcal{D}$, continuity of the relevant derivatives yields uniform constants $G_V, H_V, B_F \geq 0$ with $\|\nabla \mathcal{V}(\mathbf{z})\| \leq G_V$, $\|\nabla^2 \mathcal{V}(\mathbf{z})\| \leq H_V$, and $\|\mathbf{F}(\mathbf{z})\| \leq B_F$ for all $\mathbf{z} \in \mathcal{N}$; in particular $G_{V,k}, H_{V,k}$ may be replaced by G_V, H_V for $k \geq k_0$.

Assumption 6 (Uniform residual budget on $\mathcal{N}$). *There exist finite $b_F, \bar{b}_G, \bar{\sigma}_2, \bar{\rho}_c, \bar{R} \geq 0$ such that, for all $k \geq k_0$:*

- (a) $\bar{b}_k^F \leq \bar{b}_F$, $\bar{b}_k^G \leq \bar{b}_G$, and $\sigma_{2,k} \leq \bar{\sigma}_2$;
- (b) $\rho_{c,k} \leq \bar{\rho}_c$;
- (c) $\mathbb{E}[\varepsilon_k^{\text{tot}}] \leq \bar{R}$.

The baseline one-step and stationarity results first use exact local-model terms, $\hat{e}_k = \langle \nabla \mathcal{V}(\mathbf{z}_k), \hat{\mathbf{F}}_k \rangle$ and $\hat{c}_k = \hat{\mathbf{F}}_k^\top \nabla^2 \mathcal{V}(\mathbf{z}_k) \hat{\mathbf{F}}_k + \rho_{c,k}$. The curvature-improvement theorem then retains box constraints, local-model estimation error, and safeguard loss explicitly.

B. Expected one-step change under biased oracles

Define the second-moment field bound $M_F \triangleq 3B_F^2 + 3\bar{b}_F^2 + 3\bar{\sigma}_2^2$.

Theorem 4 (Expected one-step change in $\mathcal{V}$). *Suppose Assumptions 1, 2, 3, 4, and 6 hold, the algorithm uses the exact-autodifferentiation realization, and the iterates satisfy $\mathbf{z}_k \in \mathcal{N}$ for all $k \geq k_0$ as in Assumption 5(a). The uniform bounds G_V, H_V, M_F are then valid on $\mathcal{N}$. Then for every $k \geq k_0$ and every deterministic $\bar{\beta} \geq 0$,*

$$\begin{aligned} \mathbb{E}[\mathcal{V}(\mathbf{z}_{k+1}) - \mathcal{V}(\mathbf{z}_k) \mid \mathcal{F}_k] &\leq -\bar{\beta} \langle \nabla \mathcal{V}(\mathbf{z}_k), \mathbf{F}(\mathbf{z}_k) \rangle + \bar{\beta} G_V \bar{b}_F \\ &\quad + \frac{1}{2} \bar{\beta}^2 (H_V M_F + \bar{\rho}_c) \\ &\quad + \mathbb{E}[\varepsilon_k^{\text{tot}} \mid \mathcal{F}_k]. \end{aligned} \quad (23)$$

Proof. See Appendix A. $\square$

The reference scalar $\bar{\beta} \geq 0$ is an analytical comparator. The proof removes the curvature-oracle bias $\bar{b}_G$ from (23) by restricting the predicted decrease to the pure- $\mathbf{F}$ coordinate, which uses $\hat{e}_k$ and $\hat{c}_k$. Define the error term $\mathcal{R}(\bar{\beta}) \triangleq \bar{\beta} G_V \bar{b}_F + \frac{1}{2} \bar{\beta}^2 (H_V M_F + \bar{\rho}_c) + \bar{R}$.

C. Finite-time average stationarity and contraction

Theorem 5 (Local finite-time average stationarity). *Under the assumptions of Theorem 4, suppose additionally that Assumption 5(b) holds with $\mu_\Phi > 0$. Then, for every $\bar{\beta} > 0$ and integer $K > k_0$,*

$$\frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\Phi(\mathbf{z}_k)] \leq \frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{\bar{\beta} \mu_\Phi (K - k_0)} + \frac{\mathcal{R}(\bar{\beta})}{\bar{\beta} \mu_\Phi}. \quad (24)$$

Proof. See Appendix A. $\square$

The left side of Eq. (24) is the Cesàro, or time, average of the expected stationarity residuals. If I_K is uniform on $\{k_0, \dots, K-1\}$ and independent of the algorithm, this average equals $\mathbb{E}[\Phi(\mathbf{z}_{I_K})]$. Thus the expected residual of a uniformly selected iterate has an $O(1/(K - k_0))$ transient plus the stated error floor; the policy parameters are not averaged.

Theorems 4–5 use only $\hat{\Delta}_k^* \geq \hat{\Delta}_k^F$. Assumption 7 lower-bounds the extra decrease in the local model of $\mathcal{V}$ from the curvature step size by the stationarity residual. It applies directly to the implemented box-QP, including active caps. For fixed caps $\beta_{\max}, \gamma_{\max} > 0$, define

$$\hat{\Delta}_k^{\text{box}} \triangleq - \min_{\substack{0 \leq \beta \leq \beta_{\max} \\ 0 \leq \gamma \leq \gamma_{\max}}} \hat{q}_k(\beta, \gamma),$$

$$\hat{\Delta}_k^{F, \text{box}} \triangleq - \min_{0 \leq \beta \leq \beta_{\max}} \hat{q}_k(\beta, 0).$$

Assumption 7 (Expected model improvement from the curvature direction). *There exist $C_G > 0$, $\bar{r}_G < \infty$, and nonnegative $\mathcal{F}_k$ -measurable variables $r_{G,k}^{\text{mar}}$ such that, for every $k \geq k_0$ almost surely,*

$$\mathbb{E}[\mathcal{D}_{G,k}^{\text{box}} \mid \mathcal{F}_k] \geq C_G \Phi(\mathbf{z}_k) - r_{G,k}^{\text{mar}}, \quad \mathbb{E}[r_{G,k}^{\text{mar}}] \leq \bar{r}_G, \quad (25)$$

where $\mathcal{D}_{G,k}^{\text{box}} \triangleq \hat{\Delta}_k^{\text{box}} - \hat{\Delta}_k^{F, \text{box}}$.

The quantity $\mathcal{D}_{G,k}^{\text{box}}$ is the extra decrease in the local model of $\mathcal{V}$ obtained when the QP optimizes both step sizes instead of fixing $\gamma = 0$. The solved subproblem gives this quantity directly. Eq. (25) requires its conditional mean to grow at least as fast as the population stationarity residual Φ , up to $r_{G,k}^{\text{mar}}$. A pathwise cap-valid certificate, which implies Assumption 7 with $r_{G,k}^{\text{mar}} = 0$, follows by letting $\hat{\beta}_k^F$ be the pure-field box minimizer and setting $r_{G,k}^{\text{box}} = \hat{d}_k + \hat{b}_k \hat{\beta}_k^F$. The feasible comparator $(\hat{\beta}_k^F, \tilde{\gamma})$ gives, for $0 \leq \tilde{\gamma} \leq \gamma_{\max}$,

$$\mathcal{D}_{G,k}^{\text{box}} \geq \tilde{\gamma} r_{G,k}^{\text{box}} - \frac{1}{2} \hat{a}_k \tilde{\gamma}^2.$$

Thus, if on $\mathcal{N}$, $0 \leq \Phi \leq \bar{\Phi} > 0$, $0 < \hat{a}_k \leq \bar{a}$, and $r_{G,k}^{\text{box}} \geq m_G \sqrt{\Phi(\mathbf{z}_k)}$, choosing $\tilde{\gamma} = \min\{\gamma_{\max}, r_{G,k}^{\text{box}}/\hat{a}_k\}$ proves (25) with

$$C_G = \min \left\{ \frac{m_G^2}{2\bar{a}}, \frac{\gamma_{\max} m_G}{2\sqrt{\bar{\Phi}}} \right\}.$$

Indeed, the comparator decrease is respectively $(r_{G,k}^{\text{box}})^2/(2\hat{a}_k)$ before saturation and at least $\gamma_{\max} r_{G,k}^{\text{box}}/2$ after it. For the uncapped problem, the sharper reduced coordinates $r_{G,k} = \hat{d}_k + \hat{b}_k \hat{e}_k/\hat{c}_k$ and $h_{G,k} = \hat{a}_k - \hat{b}_k^2/\hat{c}_k$ give $\mathcal{D}_{G,k} = r_{G,k}^2/(2h_{G,k}) + [-\hat{e}_k]^2/(2\hat{c}_k) \geq r_{G,k}^2/(2h_{G,k})$ whenever the unconstrained critical point is nonnegative. Hence $|r_{G,k}| \geq m_G \sqrt{\bar{\Phi}}$ and $0 < h_{G,k} \leq \bar{h}_G$ give $C_G = m_G^2/(2\bar{h}_G)$. For the exact field-energy function and exact local-model terms, with no additional inflation, $\mathcal{V} = \phi = \frac{1}{2} \|\mathbf{F}\|^2$, $\mathbf{W}_V = \mathbf{I}$, and the pure rotation $\mathbf{F}(\mathbf{z}) = \sigma J_0(\mathbf{z} - \mathbf{z}^*)$, a cap gives $C_G = \bar{\gamma} \sigma^2 - \bar{\gamma}^2 \sigma^4/2$ with $\bar{\gamma} = \min\{\gamma_{\max}, \sigma^{-2}\}$. This expression reduces to $\bar{\gamma} - \bar{\gamma}^2/2$ when $\sigma = 1$.

Theorem 6 (Stochastic stationarity bound with curvature improvement). *Suppose Assumptions 1, 2, 3, 4, 5(a)–(b), and 7 hold, with $\mu_\Phi \geq 0$, and the uniform bounds in Assumption 6(a)–(b) hold. Suppose the accepted boxed pair satisfies $\hat{q}_k(\beta_k, \gamma_k) \leq -\hat{\Delta}_k^{\text{box}} + \varepsilon_k^{\text{step, box}}$ and*

$$\mathbb{E}[\varepsilon_k^{\text{rem}} + \varepsilon_k^{\text{model}} + \varepsilon_k^{\text{step, box}}] \leq \bar{R}_{\text{box}}.$$

If the estimation errors of Proposition 6 satisfy $\mathbb{E}[\delta_{e,k} \mid \mathcal{F}_k] \leq \bar{\delta}_e$ and $\mathbb{E}[\delta_{c,k} \mid \mathcal{F}_k] \leq \bar{\delta}_c$, define

$$\begin{aligned} \mathcal{R}_{\text{box}}(\bar{\beta}) &= \bar{\beta} (G_V \bar{b}_F + \bar{\delta}_e) + \bar{\beta}^2 (H_V M_F + \bar{\rho}_c + \bar{\delta}_c)/2 \\ &\quad + \bar{R}_{\text{box}} + \bar{r}_G. \end{aligned}$$

Then, for every deterministic $\bar{\beta} \in [0, \beta_{\max}]$ with $\bar{\beta} \mu_\Phi + C_G > 0$ and every integer $K > k_0$,

$$\begin{aligned} \frac{1}{K - k_0} \sum_{k=k_0}^{K-1} \mathbb{E}[\Phi(\mathbf{z}_k)] &\leq \frac{\mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})]}{(\bar{\beta} \mu_\Phi + C_G)(K - k_0)} \\ &\quad + \frac{\mathcal{R}_{\text{box}}(\bar{\beta})}{\bar{\beta} \mu_\Phi + C_G}. \end{aligned} \quad (26)$$

Proof. See Appendix A. $\square$

The constant C_G increases the weight multiplying Φ in the stationarity bound from $\beta\mu_\Phi$ to $\beta\mu_\Phi + C_G$. The resulting error floor is $\mathcal{R}_{\text{box}}(\beta)/(\beta\mu_\Phi + C_G)$.

Corollary 2 (Local geometric contraction). *In addition to the assumptions of Theorem 4, suppose Assumption 5(c) holds with $\mu_V > 0$. Then for every $\beta \in (0, 1/\mu_V]$ and every $k \geq k_0$,*

$$\mathbb{E}[\mathcal{V}(\mathbf{z}_k)] \leq (1 - \beta\mu_V)^{k-k_0} \mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})] + \frac{\mathcal{R}(\beta)}{\beta\mu_V}.$$

Proof. See Appendix A. $\square$

Corollary 3 (Worst-case return and Nash-gap bound). *Suppose the assumptions of Corollary 2 hold, $\mathcal{A}$ and $\mathcal{B}$ are finite, $\lambda_P > 0$, and $\mathcal{P} = \text{Gap}_\varepsilon$ in Eq. (11). For $\beta \in (0, 1/\mu_V]$, write $q_\beta = 1 - \beta\mu_V$ and $B_\varepsilon = \varepsilon(\log |\mathcal{A}| + \log |\mathcal{B}|)/(1 - \alpha)$. Then, for every $k \geq k_0$,*

$$\begin{aligned} \mathbb{E}[v^* - R_{\text{BR}}(\pi_{\theta_k})] &\leq \mathbb{E}[\text{Gap}_0(\mathbf{z}_k)] \\ &\leq \frac{\kappa_P}{\lambda_P} \left[q_\beta^{k-k_0} \mathbb{E}[\mathcal{V}(\mathbf{z}_{k_0})] + \frac{\mathcal{R}(\beta)}{\beta\mu_V} \right] + B_\varepsilon. \end{aligned} \quad (27)$$

Proof. Eq. (11) gives $\text{Gap}_\varepsilon(\mathbf{z}_k) \leq (\kappa_P/\lambda_P)\mathcal{V}(\mathbf{z}_k)$. Apply Corollary 2 and then Proposition 1. $\square$

V. LOCAL-MODEL REALIZATIONS AND IMPLEMENTATION

A. Exact local-model evaluation

The terms in (12) are assembled from one field oracle $\hat{\mathbf{F}}_k$, one curvature oracle $\hat{\mathbf{G}}_k$, one gradient $\nabla\mathcal{V}(\mathbf{z}_k)$, and the Hessian–vector products of $\mathcal{V}$ along $\hat{\mathbf{F}}_k$ and $\hat{\mathbf{G}}_k$. When a fixed differentiable inner loop implements $\mathcal{P}$, autodifferentiation passes through it; implicit differentiation handles an exact local envelope. Because the field-energy component contains $\mathbf{F}^\top \mathbf{W}_V \mathbf{F}/2$, exact Hessian–vector products of $\mathcal{V}$ generally involve third derivatives of the underlying game objective; the secant realization below avoids constructing those derivatives. For exact local-model terms, set $(\tilde{e}_k, \tilde{d}_k, \tilde{c}_k, \tilde{a}_k, \tilde{b}_k) = (e_k, d_k, c_k, a_k, b_k)$; the spectral shift then gives $(\hat{e}_k, \hat{d}_k, \hat{c}_k, \hat{a}_k, \hat{b}_k) = (e_k, d_k, c_k + \rho_{c,k}, a_k + \rho_{a,k}, b_k)$.

Proposition 5 (Exact-autodifferentiation realization). *Under the exact-autodifferentiation realization, $\hat{q}_k \equiv q_k$, hence $\varepsilon_k^{\text{model}} = 0$.*

Proof. Term-by-term comparison gives $\hat{q}_k(\beta, \gamma) = q_k(\beta, \gamma)$ for every (β, γ) ; hence $q_k(\beta_k, \gamma_k) \leq \hat{q}_k(\beta_k, \gamma_k)$ holds with $\varepsilon_k^{\text{model}} = 0$. $\square$

Exact autodifferentiation therefore removes local-model estimation error.

B. Approximate local-model estimation

For probe radii $\delta_{F,k}, \delta_{G,k} > 0$, a directional-secant realization holds $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$ fixed and evaluates $V_{\pm F} = \mathcal{V}(\mathbf{z}_k \pm \delta_{F,k}\hat{\mathbf{F}}_k)$, $V_{\pm G} = \mathcal{V}(\mathbf{z}_k \pm \delta_{G,k}\hat{\mathbf{G}}_k)$, and $V_{\pm F, \pm G} = \mathcal{V}(\mathbf{z}_k \pm$

$\delta_{F,k}\hat{\mathbf{F}}_k \pm \delta_{G,k}\hat{\mathbf{G}}_k)$. Writing $V_0 = \mathcal{V}(\mathbf{z}_k)$, the central stencils are

$$\begin{aligned} \tilde{e}_k &= \frac{V_{+F} - V_{-F}}{2\delta_{F,k}}, & \tilde{d}_k &= -\frac{V_{+G} - V_{-G}}{2\delta_{G,k}}, \\ \tilde{c}_k &= \frac{V_{+F} - 2V_0 + V_{-F}}{\delta_{F,k}^2}, & \tilde{a}_k &= \frac{V_{+G} - 2V_0 + V_{-G}}{\delta_{G,k}^2}, \\ \tilde{b}_k &= \frac{V_{+F,+G} - V_{+F,-G} - V_{-F,+G} + V_{-F,-G}}{4\delta_{F,k}\delta_{G,k}}, \end{aligned}$$

before diagonal inflation. With a shared oracle seed ζ_k and radius $\epsilon_k > 0$, a Jacobian–vector finite-difference curvature oracle is $\hat{\mathbf{G}}_k^{\text{fd}} = [\hat{\mathbf{F}}(\mathbf{z}_k + \epsilon_k\hat{\mathbf{F}}_k, \zeta_k) - \hat{\mathbf{F}}(\mathbf{z}_k, \zeta_k)]/\epsilon_k$. A moment-based realization smooths the second-order terms by exponential moving averages while preserving the subproblem.

Proposition 6 (Local-model approximation perturbation bound). *Suppose $|\hat{e}_k - e_k| \leq \delta_{e,k}$, $|\hat{d}_k - d_k| \leq \delta_{d,k}$, $|\hat{c}_k - (c_k + \rho_{c,k})| \leq \delta_{c,k}$, $|\hat{a}_k - (a_k + \rho_{a,k})| \leq \delta_{a,k}$, $|\hat{b}_k - b_k| \leq \delta_{b,k}$. Then (18) holds with*

$$\varepsilon_k^{\text{model}} = \beta_k \delta_{e,k} + \gamma_k \delta_{d,k} + \frac{1}{2} \beta_k^2 \delta_{c,k} + \beta_k \gamma_k \delta_{b,k} + \frac{1}{2} \gamma_k^2 \delta_{a,k}. \quad (28)$$

Proof. See Appendix A. $\square$

Local-model estimation errors therefore enter the stochastic residual budget.

C. Safeguarded computation

Implementations may impose box constraints through caps $\beta_{\max}, \gamma_{\max} > 0$, replacing the cone problem by $\min_{(\beta, \gamma) \in [0, \beta_{\max}] \times [0, \gamma_{\max}]} \hat{q}_k(\beta, \gamma)$. For the cone analysis, any loss relative to the cone optimum is recorded in $\varepsilon_k^{\text{step}}$; for Theorem 6, loss relative to the box optimum is $\varepsilon_k^{\text{step, box}}$. Each iteration queries $(\hat{\mathbf{F}}_k, \hat{\mathbf{G}}_k)$, recomputes the five local-model terms, inflates the diagonal if necessary, solves the two-dimensional box QP by Proposition 4, and applies (10). If the trial segment leaves a prescribed implementable trust region $\mathcal{T} \subset \mathcal{D}$, a common backtracking factor shrinks (β_k, γ_k) ; the analysis assumes the accepted trajectory lies in $\mathcal{N}$, and records lost model decrease in $\varepsilon_k^{\text{step}}$. For bounded $\mathcal{T}$, the implementation reduces the probe radii as needed until every directional-stencil point lies in $\mathcal{T}$.

VI. EXPERIMENTS

We evaluate the proposed mechanism at four levels. Section VI-A tests the skew threshold and the local-model identity; Sections VI-B–VI-C test population performance; and Section VI-D validates trajectory-sampled field and curvature oracles. For the Markov games, dynamic programming supplies the maximizing player’s unregularized worst-case return R_{BR} and exploitability Gap_0 as the primary performance metrics. The regularized gap Gap_ε and field norm $\|\mathbf{F}\|$ diagnose the optimization dynamics. The figures abbreviate these quantities as hard BR return and hard exploitability. Baselines are GDA, Adam-GDA [27], EGM, PPM-3, and the matched adaptive ablation noG ($\gamma_k = 0$). Unless noted, both players start simultaneous updates at initialization with learning-rate or step size caps of 0.03.

A. Controlled normal-form geometry

We test the mechanism on the exactly solvable normal linear field. Define

$$J_0 \triangleq \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

The field and Lyapunov function are

$$\mathbf{F}(\mathbf{z}) = \mathbf{A}\mathbf{z}, \quad \mathbf{A} = \mu\mathbf{I}_2 + \sigma J_0, \quad \mathcal{V}(\mathbf{z}) = \frac{1}{2} \|\mathbf{F}(\mathbf{z})\|^2.$$

Here $\mathbf{S} = \mu\mathbf{I}_2$, $\mathbf{W} = \sigma J_0$, and direct calculation gives

$$\begin{aligned} e &= \langle \nabla \mathcal{V}, \mathbf{F} \rangle = \mu(\mu^2 + \sigma^2) \|\mathbf{z}\|^2, \\ d &= -\langle \nabla \mathcal{V}, \mathbf{G} \rangle = (\mu^2 + \sigma^2)(\sigma^2 - \mu^2) \|\mathbf{z}\|^2. \end{aligned} \quad (29)$$

Thus $+\mathbf{G}$ decreases the field energy exactly when $|\sigma| > |\mu|$, matching Theorem 1; for this quadratic field-energy function, the two-dimensional Taylor model is exact.

Fig. 2 sweeps $\sigma/\mu \in [0, 2.5]$ and includes a pure-rotation trajectory. The numerical sweep reproduces the analytical threshold and the exact quadratic-model identity to numerical precision. Above the threshold, QP+G separates from noG. Under pure rotation, it drives the field energy to numerical precision, while the field direction gives zero first-order decrease in field energy.

B. Tabular Markov games with population oracles

We retain sequential structure but remove function approximation using one-state rock–paper–scissors (RPS), five-state CyclicControl, and eight-state FrequencyHopping; each player has three actions and tabular softmax logits. RPS is a standard multi-agent RL benchmark [28], and Appendix B specifies all games. For both the tabular and neural population experiments, the QP+G/noG controller uses the composite Lyapunov function

$$V_{\text{MG}}(\mathbf{z}) = 0.3 \frac{\frac{1}{2} \|\mathbf{F}(\mathbf{z})\|^2}{\kappa_F} + \frac{\text{Gap}_\varepsilon(\mathbf{z})}{\kappa_P}, \quad (30)$$

with normalizers frozen at initialization. Population autodifferentiation computes $\mathbf{F}$ and $\mathbf{G} = \nabla \mathbf{F} \mathbf{F}$ while each stencil recomputes both regularized BRs. The Bellman and Shapley solvers meet numerical tolerance, and Eq. (6) holds at every checkpoint.

Across ten initializations and 100 joint updates, QP+G attains higher final worst-case return and lower mean exploitability than noG in all three games. The paired sign tests remain significant after Holm correction across the prespecified three-game family [29]. Fig. 3 shows that the same ordering extends to the regularized gap and field norm, linking the performance gain to improved saddle-field optimization.

C. Neural-policy Markov games with population oracles

We train separate 4–8–3 tanh–softmax policies for both players, forming a 134-dimensional field on CyclicControl, FrequencyHopping, RoutingInterdiction, and SecurityPatrol; Appendix B specifies their rewards, transitions, and features.

For each game, $\alpha = 0.9$ and $\varepsilon = 0.03$; an exact differentiable value-system solve yields the population return, $\mathbf{F}$, and $\mathbf{G} = \nabla \mathbf{F} \mathbf{F}$. Each QP/noG query uses (30) and recomputes both soft BRs at the nine central-stencil values. For $D \in \{\mathbf{F}, \mathbf{G}\}$,

the probe is $\delta_D = \min\{10^{-2}, 10^{-3}/\max(\|D\|, 10^{-12})\}$; the common spectral shift has floor 10^{-8} . The regularized and unregularized best-response solvers meet numerical tolerance, and Eq. (6) holds at every checkpoint.

The evaluation uses ten test seeds and 60 simultaneous updates, disjoint from the screening runs. QP+G attains higher final worst-case return and lower mean exploitability than noG in all four games, and every paired 95% interval for the worst-case-return gain has a positive lower endpoint. Fig. 4 displays the three unanimous-seed games; SecurityPatrol has the same mean ordering. On CyclicControl, QP+G reduces mean final exploitability by 79.1%, the largest relative reduction among the neural-policy games. Write $\mathbf{F}_k \triangleq \mathbf{F}(\mathbf{z}_k)$ and $\mathbf{G}_k \triangleq \mathbf{G}(\mathbf{z}_k)$. For each QP+G seed–update pair, define the curvature contribution $C_k^G = \gamma_k \|\mathbf{G}_k\| / \max\{\beta_k \|\mathbf{F}_k\| + \gamma_k \|\mathbf{G}_k\|, 10^{-15}\}$ and skew ratio $R_k = \|\mathbf{W}_k \mathbf{F}_k\| / (\|\mathbf{S}_k \mathbf{F}_k\| + 10^{-15})$, where $\mathbf{S}_k = \text{sym}(\nabla \mathbf{F}(\mathbf{z}_k))$ and $\mathbf{W}_k = \text{skew}(\nabla \mathbf{F}(\mathbf{z}_k))$. The curvature coordinate activates in every game and contributes nontrivially to the update norm. The skew ratios for FrequencyHopping and RoutingInterdiction remain near unity, consistent with substantial rotational coupling. The QP selects the curvature step size from the five estimated terms in the local model of the one-step change in $\mathcal{V}$.

A disjoint screening set selects one global baseline rate from $\{0.001, 0.003, 0.01, 0.03\}$: 0.03 for GDA, EGM, and PPM-3, and 0.001 for Adam-GDA. Under this protocol, QP+G achieves the highest mean worst-case return and the lowest mean exploitability in all four games.

D. Finite-Trajectory Oracle Validation

For CyclicControl, the two 4–8–3 policies generate 128 trajectories of horizon $H = 16$ (2048 transitions) per update. We compare the final worst-case return of QP+G and noG after 60 joint updates. Detached current-policy probabilities define the behavior distribution, and one batch $\mathcal{B}_k$ evaluates all stencils using unclipped per-decision prefix ratios and the infinitely differentiable Monte Carlo estimator (DiCE) [30] for causal higher-order score terms. Let $\hat{J}_{H,k}$ denote the resulting finite-horizon DiCE return estimator for batch $\mathcal{B}_k$. We use $\hat{\mathbf{F}}_k = \text{diag}(-1, 1) \nabla \hat{J}_{H,k}$ and $\hat{\mathbf{G}}_k = \text{diag}(-1, 1) \nabla^2 \hat{J}_{H,k} \hat{\mathbf{F}}_k$, detaching $\hat{\mathbf{F}}_k$ in the Hessian–vector product. At stencil point $\mathbf{z}$, the controller uses $\hat{V}_k(\mathbf{z}) = \|\hat{\mathbf{F}}_k(\mathbf{z}; \mathcal{B}_k)\|^2 / (2\hat{\kappa}_F)$, with $\hat{\kappa}_F$ frozen from an independent initialization batch. Training evaluates this field-energy function directly. Population quantities are used only for checkpoint evaluation. For $\mathcal{V}(\mathbf{z}) = \|\mathbf{F}(\mathbf{z})\|^2 / (2\kappa_F)$, Assumption 2 models the field and curvature oracles, while dependent same-batch stencil errors enter the local-model estimation-error budget of Proposition 6 and Theorem 6.

On ten held-out test seeds, QP+G improves final worst-case return and reduces mean exploitability by 54.0% relative to noG; the paired 95% interval for the return gain has a positive lower endpoint. QP+G also lowers the final population field norm. The curvature coordinate remains active at every update, confirming that the sampled-oracle result uses both adaptive directions.

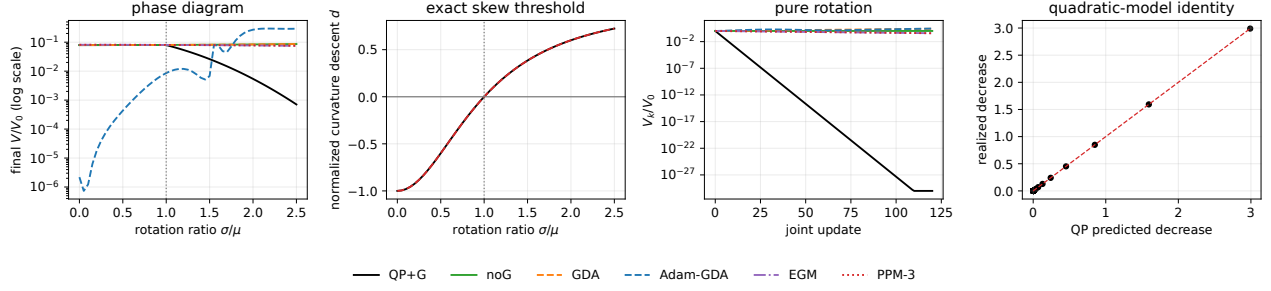


Fig. 2. Controlled normal-form geometry: normalized field energy, the analytical and computed curvature decrease term d , pure-rotation trajectories, and predicted versus realized QP decrease. The vertical line marks $\sigma/\mu = 1$.

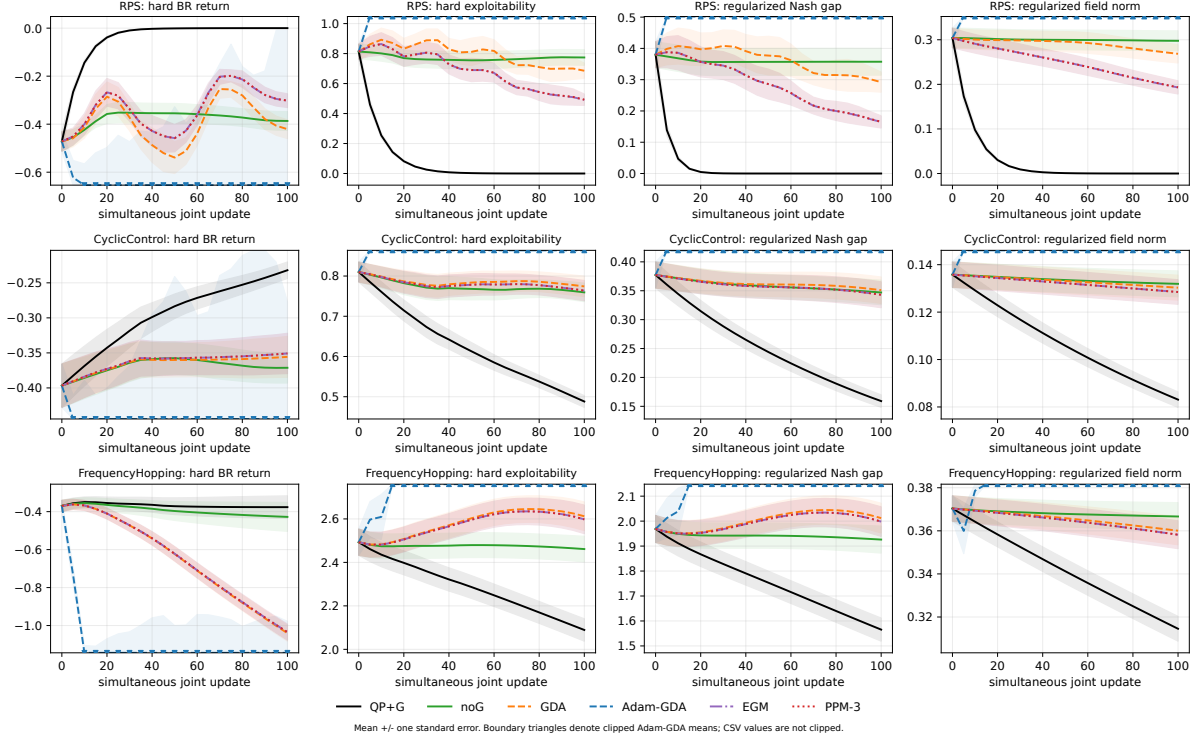


Fig. 3. Tabular Markov games with population oracles (mean $\pm$ standard error, ten seeds). Columns show worst-case return, exploitability, regularized gap, and field norm; triangles denote clipped Adam-GDA means.

CONCLUSION

We introduced an online controller for the field and curvature step sizes in stochastic zero-sum Markov-game policy optimization. The Jacobian identity states when the curvature direction decreases field energy, and the curvature-improvement condition measures its extra decrease in the local model of $\mathcal{V}$. The QP never increases this model relative to feasible field-only and fixed-coupling step-size pairs. Our finite-time bounds control the average stationarity residual and $\mathbb{E}[\mathcal{V}]$; the performance corollary then bounds the unregularized Nash gap and worst-case-return loss. In the tested games, QP+G improves worst-case return and reduces exploitability.

APPENDIX A PROOFS

a) Proof of Proposition 3: The GDA identity is immediate. On $\bar{B}(\mathbf{z}; r) \subset \mathcal{D}$, let M_i bound $\nabla^i \mathbf{F}$ for $i = 1, 2$ and B_F

bound $\mathbf{F}$. For $s \leq \min\{r/B_F, (2M_1)^{-1}, 1\}$, $T_s(\mathbf{u}) = \mathbf{z} - s\mathbf{F}(\mathbf{u})$ is a contracting self-map; therefore, its PPM fixed point has $\Delta = \mathbf{z}_+ - \mathbf{z}$ and $\|\Delta\| \leq sB_F$. Taylor expansion gives $\Delta + s\mathbf{F} = -s\nabla\mathbf{F}\Delta - s\mathbf{r}_2$, $\|\mathbf{r}_2\| \leq M_2\|\Delta\|^2/2$, hence $\Delta = -s\mathbf{F} + s^2\nabla\mathbf{F}\mathbf{F} + O(s^3)$ uniformly. For EGM, $\mathbf{F}(\mathbf{z} - s\mathbf{F}) = \mathbf{F} - s\nabla\mathbf{F}\mathbf{F} + O(s^2)$ uniformly; multiplication by $-s$ gives the same expansion. ■

b) Proof of Lemma 1: With $\mathbf{A} = \nabla\mathbf{F}$ and $\nabla\phi = \mathbf{A}^\top\mathbf{F}$, $\langle\nabla\phi, -\mathbf{F}\rangle = -\mathbf{F}^\top\mathbf{S}\mathbf{F}$ and $\langle\nabla\phi, \mathbf{G}\rangle = \mathbf{F}^\top\mathbf{A}^2\mathbf{F}$. Moreover, $\mathbf{A}^2 = \mathbf{S}^2 + \mathbf{S}\mathbf{W} + \mathbf{W}\mathbf{S} + \mathbf{W}^2$. The cross term vanishes because $\mathbf{S}\mathbf{W} + \mathbf{W}\mathbf{S}$ is antisymmetric, while $\mathbf{F}^\top\mathbf{S}^2\mathbf{F} = \|\mathbf{S}\mathbf{F}\|^2$ and $\mathbf{F}^\top\mathbf{W}^2\mathbf{F} = -\|\mathbf{W}\mathbf{F}\|^2$, proving (7)–(8). ■

c) Proof of Theorem 1: Eq. (8) proves (9). For (a), $\mathbf{S}\mathbf{F} = 0$ gives $\langle\nabla\phi, \mathbf{G}\rangle = -\|\mathbf{W}\mathbf{F}\|^2 < 0$ when $\mathbf{W}\mathbf{F} \neq 0$, whereas (7) gives $\langle\nabla\phi, -\mathbf{F}\rangle = 0$. For (b), $\mathbf{W} = 0$ gives $\langle\nabla\phi, \mathbf{G}\rangle = \|\mathbf{S}\mathbf{F}\|^2 \geq 0$. For (c), normality on the invariant plane gives $\mathbf{S} = \mu\mathbf{I}$ and $\mathbf{W} = \sigma\mathbf{J}_0$; hence (9) is equivalent to $|\sigma| > |\mu|$. ■

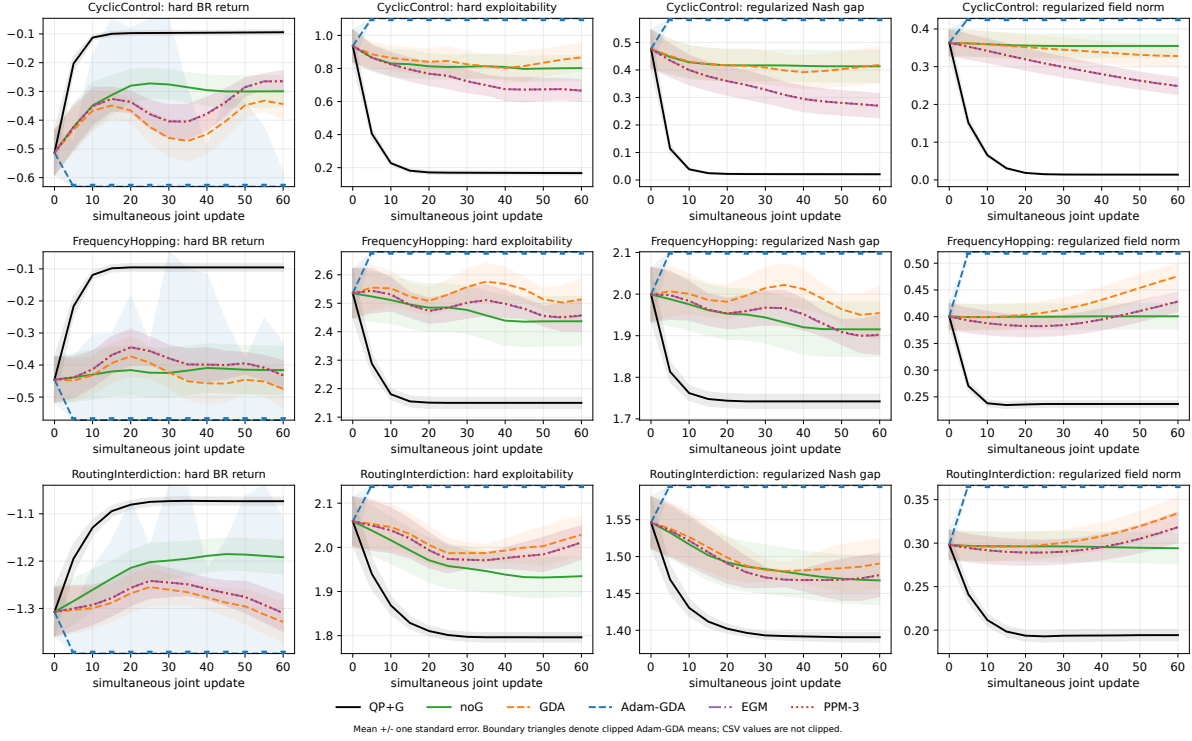


Fig. 4. Neural-policy Markov games for the three unanimous-seed QP+G/noG comparisons (mean $\pm$ standard error, ten seeds). Columns match Fig. 3; triangles denote clipped Adam-GDA means.

d) *Proof of Corollary 1:* For the linear field $F(z) = A(z - z^*)$, $\nabla F(z) = A$ and $G(z) = AF(z) = A^2(z - z^*)$. Collinearity $G = \lambda F$ is equivalent to $AF = \lambda F$. If $A^\top = -A$ and $A^2 = -\sigma^2 I$ on the relevant subspace, then $F = Ae \perp e$ and $G = -\sigma^2 e$; hence $F \perp G$, $\langle \nabla \phi, -F \rangle = 0$, and $\langle \nabla \phi, G \rangle = -\sigma^2 \|F\|^2 = -\sigma^4 \|e\|^2$. ■

e) *Proof of Lemma 2:* Nonnegative summands give $\mathcal{V} \geq 0$, and $\mathcal{V} = 0$ forces $\mathcal{E}_F = 0$, hence $F = 0$ because $\lambda_F > 0$ and $W_V \succ 0$. Smoothness and the gradient formula follow by composition and the chain rule; $\mathcal{V} \geq \lambda_F \mathcal{E}_F / \kappa_F$ with $\Phi = 2\mathcal{E}_F$ gives (d). ■

f) *Proof of Lemma 3:* Taylor's theorem along the admissible segment gives the constant, linear, and quadratic terms in (13) and remainder $\nabla^3 \mathcal{V}(z_k + \theta \Delta_k) [\Delta_k]^3 / 6$ for some $\theta \in (0, 1)$. Substitution of (12) and $\|\Delta_k\| \leq \beta \|\hat{F}_k\| + \gamma \|\hat{G}_k\|$ gives the bound. ■

g) *Proof of Theorem 2:* Positive definiteness makes $\hat{q}_k$ strictly convex. Solving $\hat{H}_k(\beta, \gamma)^\top = (\hat{e}_k, \hat{d}_k)^\top$ gives (15); the explicit inverse gives (16). If infeasible, the Karush–Kuhn–Tucker (KKT) point lies on an axis, whose clipped scalar minimizers are the stated candidates. ■

h) *Proof of Theorem 3:* The three comparator sets are subsets of $\mathbb{R}_+^2$, proving dominance by set inclusion; their formulas follow by scalar minimization. Completing the square in (16) gives $\hat{\Delta}_k^* = \hat{e}_k^2 / (2\hat{c}_k) + (\hat{b}_k \hat{e}_k + \hat{c}_k \hat{d}_k)^2 / (2\hat{c}_k \hat{\delta}_k)$. Subtracting $[\hat{e}_k]_+^2 / (2\hat{c}_k)$ proves (17) and its equality condition. ■

i) *Proof of Lemma 4:* Taylor's bound and nonnegative stabilizers give $\mathcal{V}(z_{k+1}) - \mathcal{V}(z_k) \leq q_k + \varepsilon_k^{\text{rem}}$. Applying

successively the definitions of $\varepsilon_k^{\text{model}}$ and $\varepsilon_k^{\text{step}}$ gives (19). ■

j) *Proof of Lemma 5:* The step caps and $(a+b)^3 \leq 4(a^3 + b^3)$ imply $\varepsilon_k^{\text{rem}} \leq (2M_{3,k}/3)(\beta_{\max} + \gamma_{\max})^3 (\|\hat{F}_k\|^3 + \|\hat{G}_k\|^3)$. Conditional expectation gives (20); tower expectation gives uniformity. ■

k) *Proof of Lemma 6:* Lemma 2(c) and cancellation of the antisymmetric quadratic form give a field-energy contribution $(\lambda_F / \kappa_F) F^\top \text{sym}(W_V \nabla F) F$. Applying (21)–(22) yields $\langle \nabla \mathcal{V}, F \rangle \geq (\lambda_F m_F / \kappa_F - \delta_P - \delta_C) \Phi$. ■

l) *Proof of Theorem 4:* Fix $k \geq k_0$ and $\bar{\beta} \geq 0$. Theorem 3 and $[u]_+^2 / (2v) \geq \bar{\beta} u - \bar{\beta}^2 v / 2$ give $\hat{\Delta}_k^* \geq \bar{\beta} \hat{e}_k - \bar{\beta}^2 \hat{c}_k / 2$. Lemma 4 then gives $\Delta \mathcal{V} \leq -\bar{\beta} \hat{e}_k + \bar{\beta}^2 \hat{c}_k / 2 + \varepsilon_k^{\text{tot}}$. Conditional expectation and Assumption 2 yield $-\mathbb{E}[\bar{\beta} \hat{e}_k \mid \mathcal{F}_k] \leq -\bar{\beta} \langle \nabla \mathcal{V}, F \rangle + \bar{\beta} G_V \bar{b}_F$. Moreover, $\hat{c}_k \leq H_V \|\hat{F}_k\|^2 + \rho_{c,k}$ and the three-term square bound give $\mathbb{E}[\hat{c}_k \mid \mathcal{F}_k] \leq H_V M_F + \bar{\rho}_c$. Combining proves (23). ■

m) *Proof of Theorem 5:* Taking total expectation in (23) and using Assumptions 6(c) and 5(b) gives $\mathbb{E}[\Delta \mathcal{V}] \leq -\bar{\beta} \mu_\Phi \mathbb{E}[\Phi(z_k)] + \mathcal{R}(\bar{\beta})$. Sum from k_0 to $K-1$, use $\mathcal{V} \geq 0$, and divide by $\beta \mu_\Phi (K - k_0)$. ■

n) *Proof of Theorem 6:* Taylor's bound, model consistency, and accepted-step suboptimality give $\Delta \mathcal{V} \leq -\hat{\Delta}_k^{\text{box}} + \varepsilon_k^{\text{rem}} + \varepsilon_k^{\text{model}} + \varepsilon_k^{\text{step, box}}$. Since $(\bar{\beta}, 0)$ is box-feasible, $\hat{\Delta}_k^{\text{box}} \geq \bar{\beta} \hat{e}_k - \bar{\beta}^2 \hat{c}_k / 2$. Thus $\Delta \mathcal{V} \leq -\bar{\beta} \hat{e}_k + \bar{\beta}^2 (\hat{c}_k + \rho_{c,k}) / 2 - \mathfrak{D}_{G,k}^{\text{box}} + \bar{\beta} \delta_{e,k} + \bar{\beta}^2 \delta_{c,k} / 2 + \varepsilon_k^{\text{box}}$, where $\varepsilon_k^{\text{box}}$ sums the three residuals above. Conditional expectation bounds the first two terms exactly as in Theorem 4, while Assumption 7 bounds the gain term by $-C_G \Phi(z_k) + r_{G,k}^{\text{mar}}$. The stated error, margin-residual,

and safeguard budgets and Assumption 5(b) therefore give $\mathbb{E}[\Delta V] \leq -(\bar{\beta}\mu_\Phi + C_G)\mathbb{E}[\Phi(\mathbf{z}_k)] + \mathcal{R}_{\text{box}}(\bar{\beta})$. Telescoping and $V \geq 0$ prove (26). ■

o) Proof of Corollary 2: Taking total expectations in (23) and applying Assumptions 5(c) and 6 gives $\mathbb{E}[V(\mathbf{z}_{k+1})] \leq (1 - \bar{\beta}\mu_V)\mathbb{E}[V(\mathbf{z}_k)] + \mathcal{R}(\bar{\beta})$. Since $q = 1 - \bar{\beta}\mu_V \in [0, 1)$, iteration yields $\mathbb{E}[V(\mathbf{z}_k)] \leq q^{k-k_0}\mathbb{E}[V(\mathbf{z}_{k_0})] + \mathcal{R}(\bar{\beta})\sum_{j \geq 0} q^j$; using $\sum_{j \geq 0} q^j = (\bar{\beta}\mu_V)^{-1}$ proves the claim. ■

p) Proof of Proposition 6: Subtracting the two quadratic models and applying the five estimation-error bounds termwise gives exactly (28). ■

APPENDIX B

FINITE MARKOV-GAME ENVIRONMENTS

All reported environments have actions $a, b \in \{0, 1, 2\}$ and initial state distribution uniform on $\{0, \dots, S-1\}$. For target map $T(s, a, b)$ and noise mass ξ , the kernel is $P(s' | s, a, b) = (1-\xi)\mathbf{1}_{\{s'=T(s,a,b)\}} + \xi\mathbf{1}_{\{s' \neq T(s,a,b)\}}/(S-1)$. Let $\varphi_s = 2\pi s/S$ and define

$$M \triangleq \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}.$$

The one-state RPS game uses reward M_{ab} and a deterministic self-loop; the remaining games are: *CyclicControl*: $S = 5$, $\xi = 0.02$, $T = (s+1 + ((a-b) \bmod 3)) \bmod S$, and $r_s(a, b) = (1 + 0.12 \cos \varphi_s)M_{\pi_s(a), \pi_s(b)}$, where $\pi_s(j) = (j-s) \bmod 3$. *FrequencyHopping*: $S = 8$, $\xi = 0.08$, $T = (s+1+a-b) \bmod S$, $q_s(a) = 0.25 \cos(\varphi_s + 2\pi a/3)$, and $r_s(a, b) = q_s(a) + 0.55$ if $a \neq b$, while $r_s(a, a) = q_s(a) - 1.10$. *RoutingInterdiction*: $S = 6$, $\xi = 0.10$, $T = (s+1 + \mathbf{1}_{\{a=b\}} + a) \bmod S$, $c_s(a) = 1 + 0.20 \sin(\varphi_s + 2\pi a/3)$, and $r_s(a, b) = c_s(a) - 0.55 - 1.35\mathbf{1}_{\{a=b\}} - 0.15\mathbf{1}_{\{a \neq b\}}$. *Security-Patrol*: $S = 8$, $\xi = 0.08$, $T = (s+1+b + \mathbf{1}_{\{a \neq b\}}) \bmod S$, $v_s(b) = 0.8 + 0.25 \cos(\varphi_s + 2\pi b/3)$, and $r_s(a, b) = v_s(b)$ if $a = b$, while $r_s(a, b) = -0.55v_s(b)$ otherwise. Neural policies use $x_s = (\sin \varphi_s, \cos \varphi_s, u_s, 1)$, where, in the order above, u_s is $s/(S-1)$, the range of q_s , the standard deviation of c_s , and the mean of v_s . The last two are grid-constant architecture coordinates; state variation is encoded by $(\sin \varphi_s, \cos \varphi_s)$.

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